

CALRIG · BUILD PLAN

CalRig prototype build plan

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DOCUMENT CONTROL

Document ID	CLR-BLD-001	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/05-build-plan.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the plan (front badge with name plate and status light and its wiring, printed door border, door panel pull handle, large heads in the first checks); new section 3.15, Figures 23 and 24 and step 12; figures and steps renumbered
0.1	2026-10-01	Amish Chadha	First build plan, with pictures by component and step; design made constructable (CLR-DDR-003)

CalRig prototype build plan

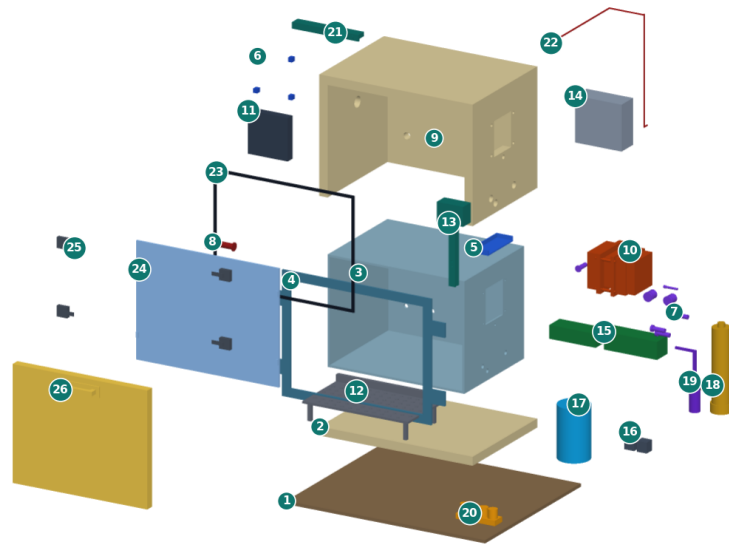
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

CalRig prototype: every component, pulled apart

Numbered in build order. Seen from the front right and above

- 1 Base plate
- 2 Jacket bottom panel
- 3 Chamber shell
- 4 Front frame
- 5 Drip tray
- 6 Fan spacers (4)
- 7 Bulkheads and cable glands
- 8 Aerosol port and valve
- 9 Jacket back, side and top panels
- 10 Peltier assembly
- 11 Mixing fan
- 12 Sensor tray
- 13 Reference mast and cluster
- 14 HEPA unit
- 15 Controller and power supply
- 16 Air pumps (2) and suction line
- 17 Bubbler
- 18 Dryer socket and column
- 19 Drain line and bottle
- 20 Jar rack and salt jars
- 21 Front badge, name plate and status light
- 22 Status light lead
- 23 Door gasket
- 24 Door with printed border and keepers
- 25 Draw latches (4)
- 26 Door panel with pull handle



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Figure 1. Every component pulled apart and numbered in build order, seen from the front right and above.

The prototype is one CalRig on a bench: a 36 L clear acrylic box wrapped in foam board, with a door at the front, a heat pump through its right wall, and the conditioning parts, the HEPA unit and the electronics standing on a plywood base around it. Figure 1 shows the 26 components in the order you make or fit them. Eleven are made in a workshop or maker space: the base plate, the five jacket panels, the chamber shell with its welded front frame, drip tray and fan spacers, the sensor tray, the door with its printed border and the door panel from sheet, and four small printed parts (the reference mast, the dryer socket, the jar rack and the front badge), with a printed pull handle on the door panel. The rest are bought and fitted: the heat pump, fans, pumps, bubbler, dryer column, HEPA unit, bulkhead fittings and cable glands, latches, sensors, the front status light and its lead, controller and power supply. The work is laser cutting and solvent welding acrylic, cutting foam board, a little drilling and printing, and wiring bought modules together with screw terminals. The parts cost about \$449, from the bill of materials.

Safety: The rig runs from a certified external 12 V supply; no mains wiring is inside it. The heat pump's outer sink reaches 50 °C and its hot side can reach 70 °C; the heaters and heat pump are cut off by two bimetal switches whatever the software does. Test smoke contains fine particles and some carbon monoxide: light the incense outside the chamber in a ventilated room and clear the chamber through the HEPA unit before opening the door. Soda lime is corrosive and lithium chloride is harmful if swallowed: wear gloves and eye protection. Acrylic solvent cement and laser-cut acrylic give off fumes: work in a ventilated space.

■ 2. What changed to make it buildable

The concept showed what the rig does; some of its parts could not be made or fixed as drawn. Each change below keeps what the rig does. The construction changes are recorded in decision record CLR-DDR-003, accepted by Amish on 2026-10-02; the last three rows are details adopted on the same day (design decisions register).

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Door	A door the size of the shell, sealing on the 6 mm wall edges; two latches with nothing to catch	A welded front frame with a 25 mm gasket face, a 430 x 330 mm door and four draw latches on frame tabs (Figures 5 and 27)	The gasket needs a face to seal on and the latches something to pull against
Door panel	A full-front foam panel standing loose in front of the latches	A 406 x 330 mm panel on the door, between the latches, on four hook-and-loop pads (Figure 22)	Held in place and off by hand
Heat pump	No clamping to the wall	Both sink bases clamp the wall with four screws through rigid sleeves in the foam (Figure 14)	Standard mounting for a through-wall heat pump
Mixing fan	Floating off the back wall	Four welded spacers, fan 15 mm off the wall (Figure 15)	No hole through the wall
Sensor tray and reference mast	Thin posts; the mast floated behind the tray	10 mm legs; a printed mast standing on the floor against the tray's rail (Figure 18)	Both stand on the floor and come out through the door
Air loops	Lines ending at the foam with no way through the wall; no outlet for the pumps; the HEPA unit not connected	Five 12 mm bulkhead fittings (wet inlet, dry inlet, suction, two to the HEPA unit) and two pumps on the base (Figures 11 to 13 and 25)	A closed loop needs a way out of the chamber as well as in
Sensor leads	One gland in the door	Two cable glands in the back wall (Figure 13)	Leads stay put when the door opens
Drip tray and drain	Named in the concept but not drawn	A tray under the inner sink, a drain fitting and a bottle on the base (Figure 9)	Condensate at cold points leaves the chamber
Base plate	12 mm plywood or HDPE	9 mm sealed birch plywood	Keeps the rig under the 14 kg limit after the added parts (13.9 kg)
Status light and name plate	Not drawn	A printed badge on the front of the jacket's top panel, its lip resting on the front frame, carrying the name plate and a 12 V panel light wired to the controller (Figures 23 and 24)	Shows that a run is going without opening the rig; the badge adds 12 mm to the height (383 mm)
Door border	Not drawn	A 22 mm black printed border on the door's front face, cut round the keepers (Figure 21)	Hides the frame edges and gasket
Door panel handle	Nothing to pull it off by	A printed pull handle glued to the panel's front (Figure 22)	The panel comes off for every warm, dry point

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Left" and "right" are as seen standing at the front of the rig, looking at the door. Workshop tolerance is 0.5 mm unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Base plate

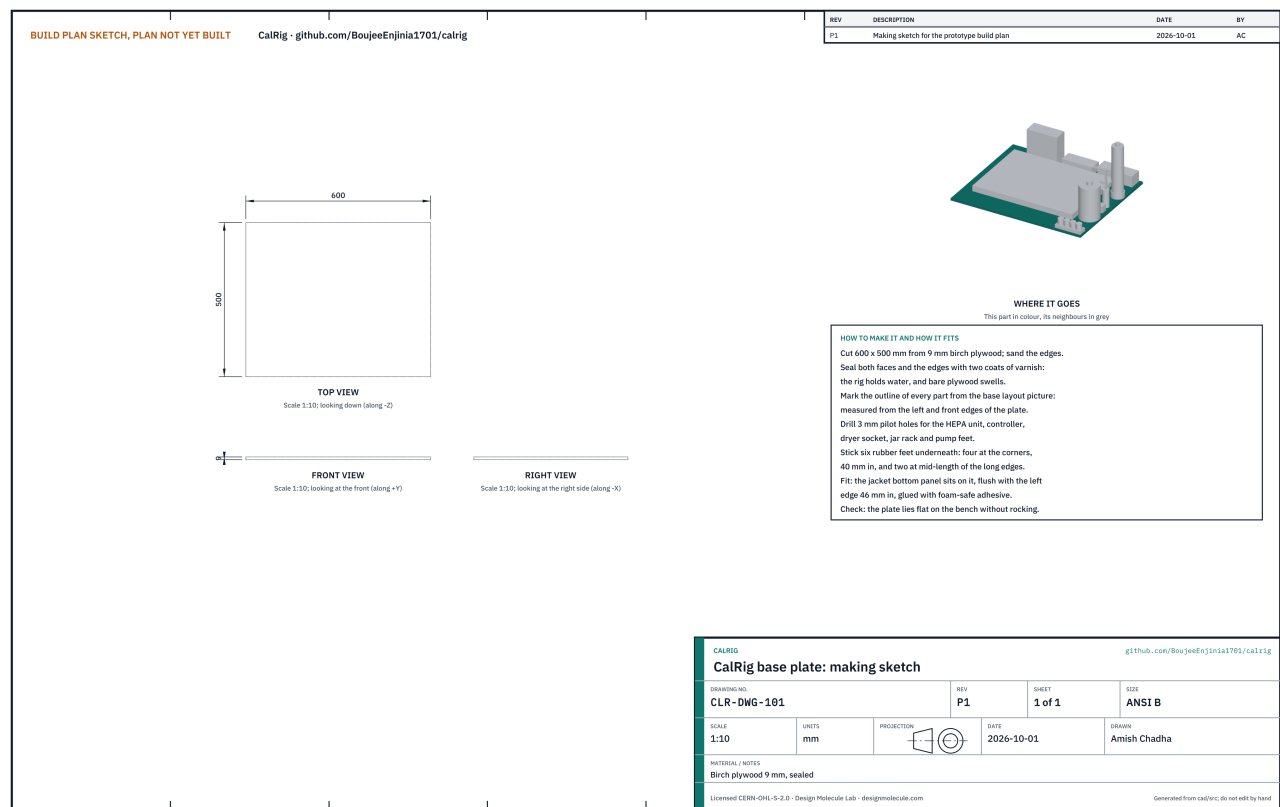
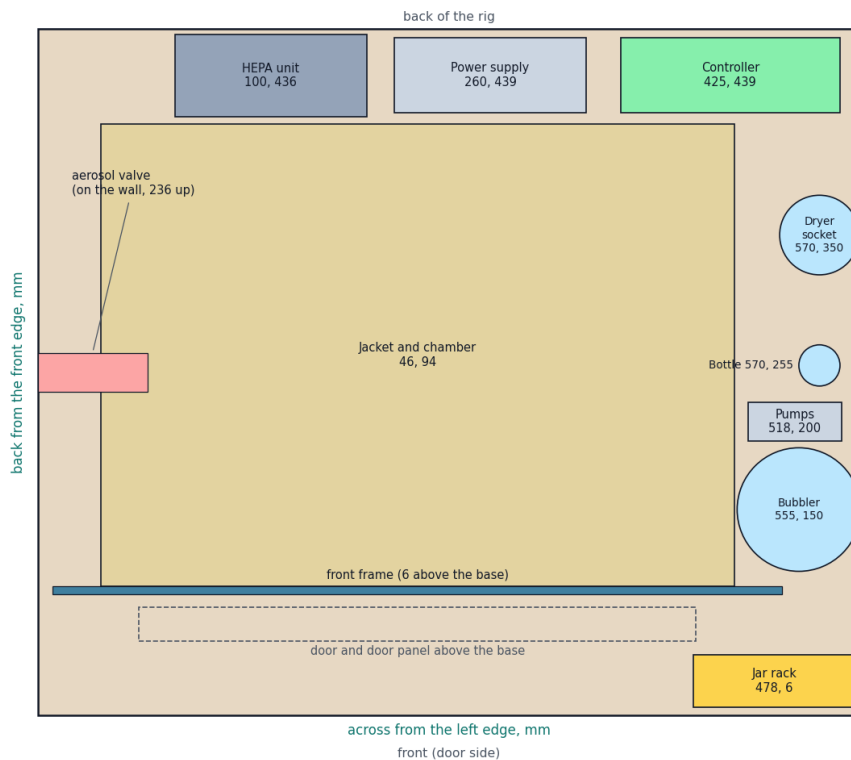


Figure 2. Base plate making sketch (CLR-DWG-101).

Base plate: where each part stands

Seen from above, front at the bottom. Figures under each name: its left and front edges (rectangles) or its centre (circles), in mm from the base plate's left and front edges.



Fixings (3 mm pilot holes):

Jacket bottom: foam-safe adhesive
 HEPA unit: four screws through its feet
 Controller: four screws through its case flanges
 Power supply: a hook-and-loop strap through two 25 mm slots
 Pumps: two screws each through their rubber mounts
 Dryer socket: three screws
 Jar rack: two screws
 Bubbler: a hook-and-loop pad under the jar; held also by its line
 Bottle: a hook-and-loop pad
 Six rubber feet underneath.

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Figure 3. Where each part stands, measured from the base plate's left and front edges.

What it is and what it is made from. The board everything stands on. Birch plywood 9 mm thick, cut to 600 x 500 mm.

How to make it.

1. Cut the blank to 600 x 500 mm, square, and sand the edges.
2. Seal both faces and all edges with two coats of varnish. The rig holds water, and bare plywood swells.
3. Mark the outline of every part from Figure 3, measured from the left and front edges.
4. Drill 3 mm pilot holes for the screws of the HEPA unit, controller, pumps, dryer socket and jar rack, through the marked feet or flanges of each part.
5. Cut two 25 mm slots for the power supply's strap, either side of its outline.
6. Stick six rubber feet underneath: four at the corners, 40 mm in, and two at mid-length of the long edges.

How it fits the parts next to it. The jacket's bottom panel is glued flat on it, 46 mm from the left edge and 94 mm from the front edge. Everything else stands on it and is screwed, strapped or padded down as Figure 3 lists.

Check before moving on. The plate lies flat on the bench without rocking.

3.2 Insulation jacket

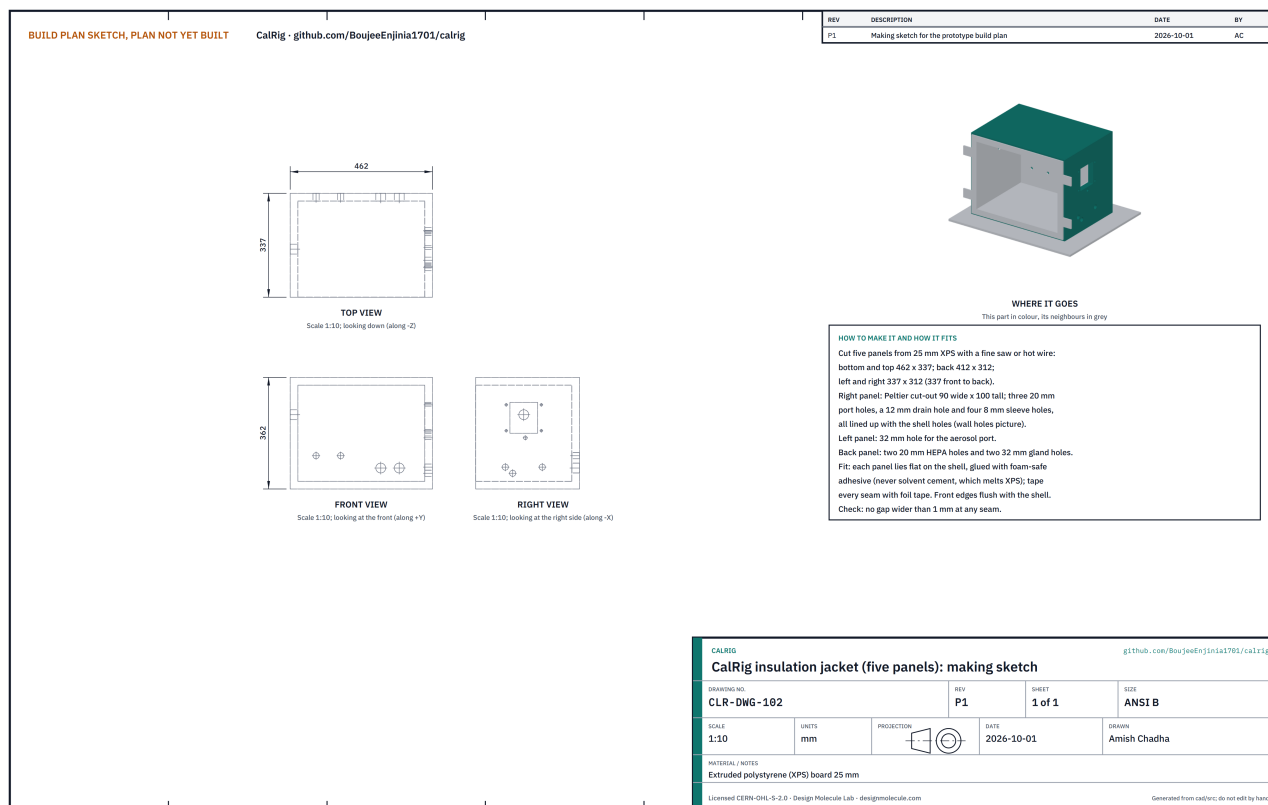


Figure 4. Insulation jacket making sketch (CLR-DWG-102). The hole positions are in Figure 6.

What it is and what it is made from. Five panels of 25 mm extruded polystyrene board that wrap the shell on every face but the front. One 1.2 x 1.2 m sheet is enough.

How to make it.

1. Cut the bottom and top panels 462 x 337 mm, the back panel 412 x 312 mm, and the left and right panels 337 mm front to back by 312 mm tall. A fine-toothed saw or a hot wire gives a clean edge.
2. Right panel: cut the heat pump opening 90 wide by 100 tall, three 20 mm holes for the air fittings, a 12 mm hole for the drain and four 8 mm holes for the clamp sleeves, at the positions of Figure 6 plus 6 mm up (the side panels sit on the bottom panel).
3. Left panel: a 32 mm hole for the aerosol port.
4. Back panel: two 20 mm holes for the HEPA fittings and two 32 mm holes for the cable glands, at the positions of Figure 6 plus 6 mm along and 6 mm up.
5. Hold each panel against the shell (section 3.3) to check that every hole lines up before gluing anything.

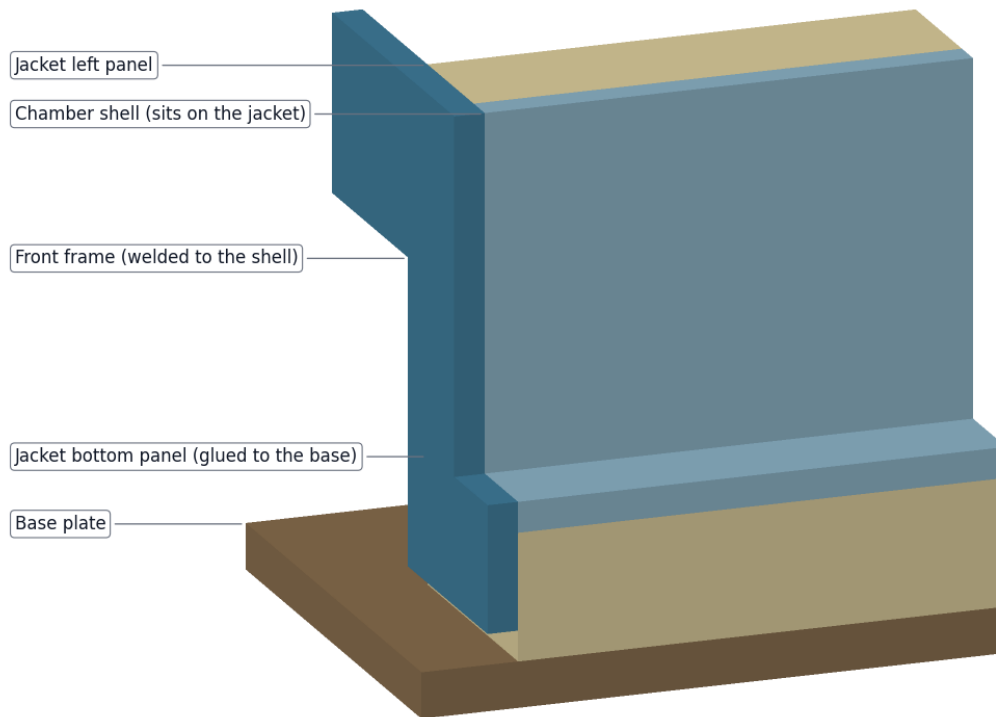
How it fits the parts next to it. The bottom panel is glued to the base; the shell sits on it; the other four panels lie flat against the shell, glued with foam-safe adhesive (never acrylic solvent cement, which melts the foam). Every seam is taped with foil tape. The front edges are flush with the front of the shell, and the front frame covers the inner 19 mm of them (Figure 5).

Check before moving on. No gap wider than 1 mm at any seam; every hole is clear of the fitting that passes through it.

3.3 Chamber shell, with its fan spacers

Joint 1: shell, jacket, frame and base (front left corner)

Cut open, seen from inside the chamber toward the left wall. The shell sits on the jacket; the frame covers their front edges



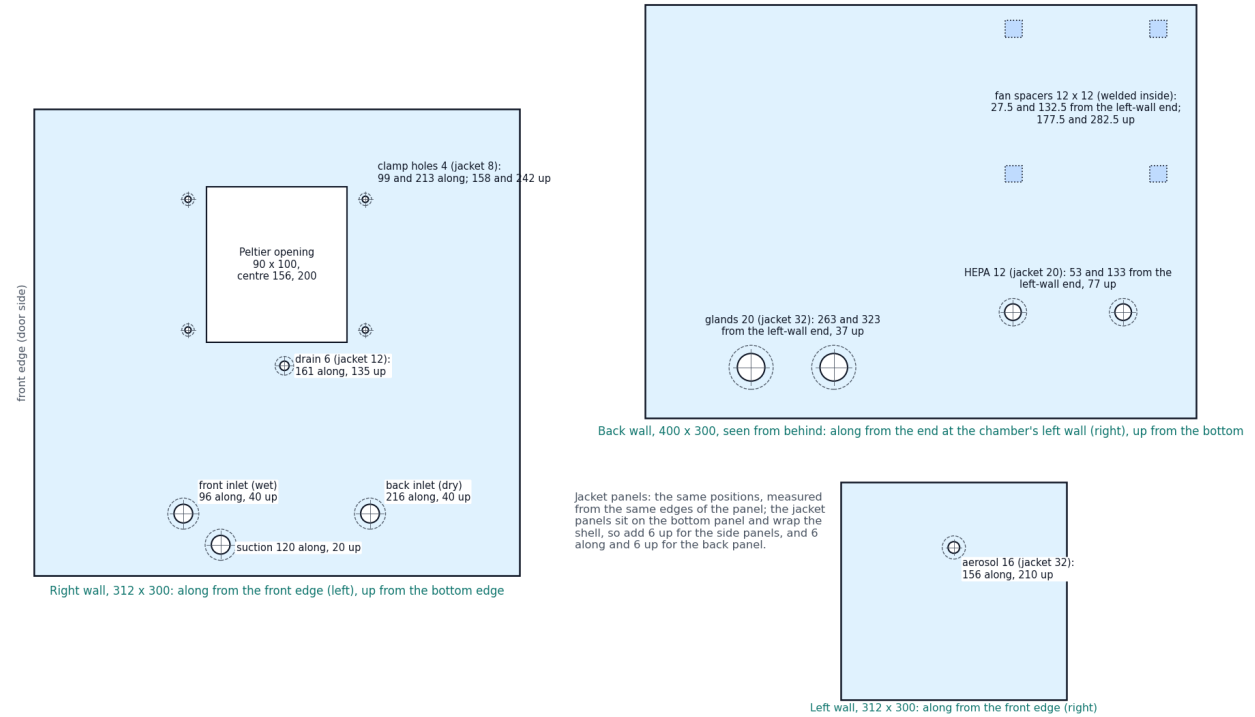
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Figure 5. The front left corner cut open: the shell sits on the jacket's bottom panel, which is glued to the base; the front frame is welded to the shell and covers the jacket's front edges.

Chamber walls: hole positions (as cut, seen from outside the chamber)

Sizes in mm from the model. Solid outline: hole in the 6 mm acrylic wall. Dashed: the larger clearance hole in the 25 mm jacket panel in the same place.



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Figure 6. Hole positions in the right, back and left walls as cut, seen from outside, with the larger jacket holes dashed.

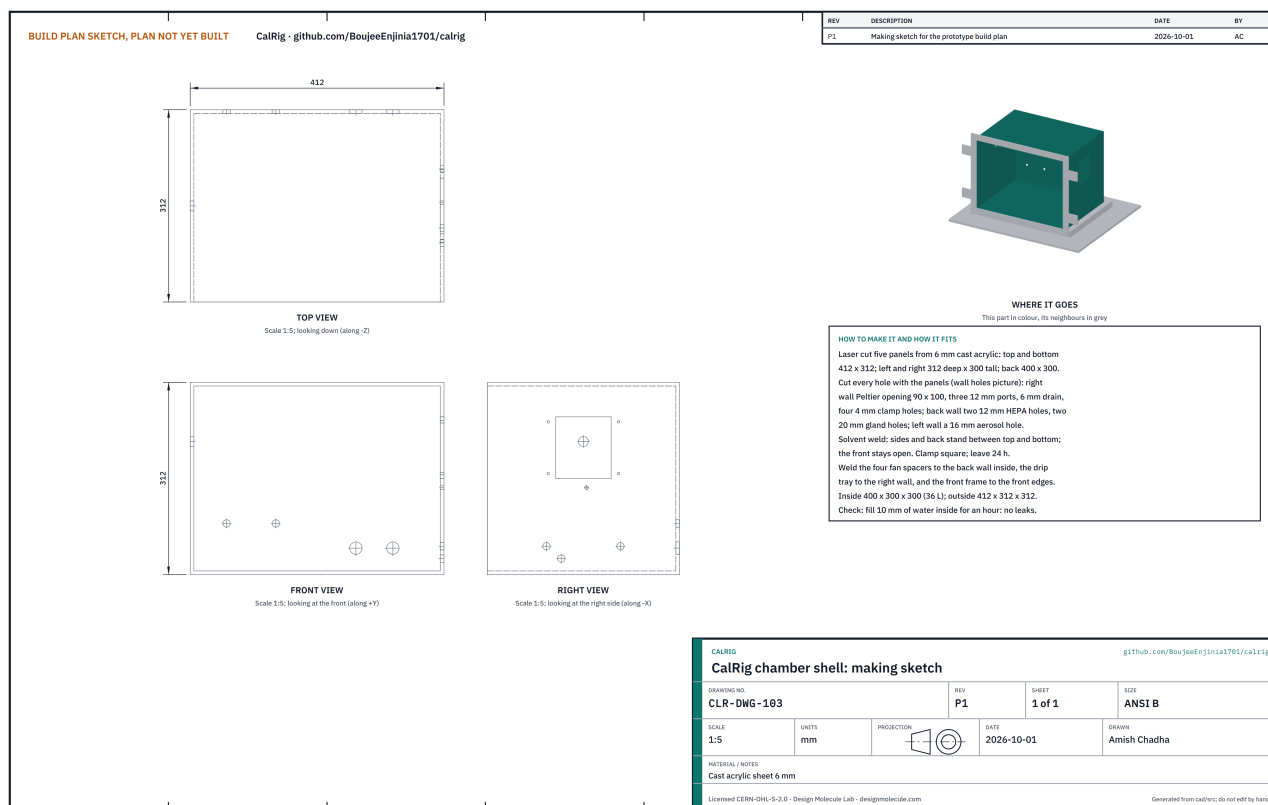


Figure 7. Chamber shell making sketch (CLR-DWG-103).

What it is and what it is made from. The sealed five-sided box the sensors sit in, 400 x 300 x 300 mm inside (36 L), open at the front. Cast acrylic sheet 6 mm thick; cast, not extruded, because it welds and machines without crazing.

How to make it.

1. Laser cut five panels: top and bottom 412 x 312 mm; left and right 312 mm deep by 300 mm tall; back 400 x 300 mm.
2. Cut every hole with its panel, at the positions of Figure 6. Right wall: the heat pump opening 90 wide by 100 tall, centred 156 from the front edge and 200 up; three 12 mm holes for the wet inlet, dry inlet and suction; a 6 mm drain hole; four 4 mm clamp holes. Back wall: two 12 mm holes for the HEPA fittings and two 20 mm holes for the cable glands. Left wall: a 16 mm hole for the aerosol port.
3. Cut four fan spacers 12 x 12 x 15 mm from offcut and drill and tap each M4 down its length.
4. Solvent weld the box: the sides and the back stand on the bottom panel, between the sides, and the top sits on them. The front stays open. Clamp it square with corner blocks and leave it 24 hours.
5. Weld the four spacers to the inside of the back wall at the positions of Figure 6.
6. Weld the drip tray (section 3.5) and the front frame (section 3.4) when they are made.

How it fits the parts next to it. The bottom sits flat on the jacket's bottom panel, glued with foam-safe adhesive (Figure 5). Every fitting seals on the acrylic with an O-ring outside and a nut inside; the jacket holes are clear of them.

Check before moving on. Stand the box on its back and pour in 10 mm of water: after an hour there is no leak at any weld. Dry it fully.

3.4 Front frame

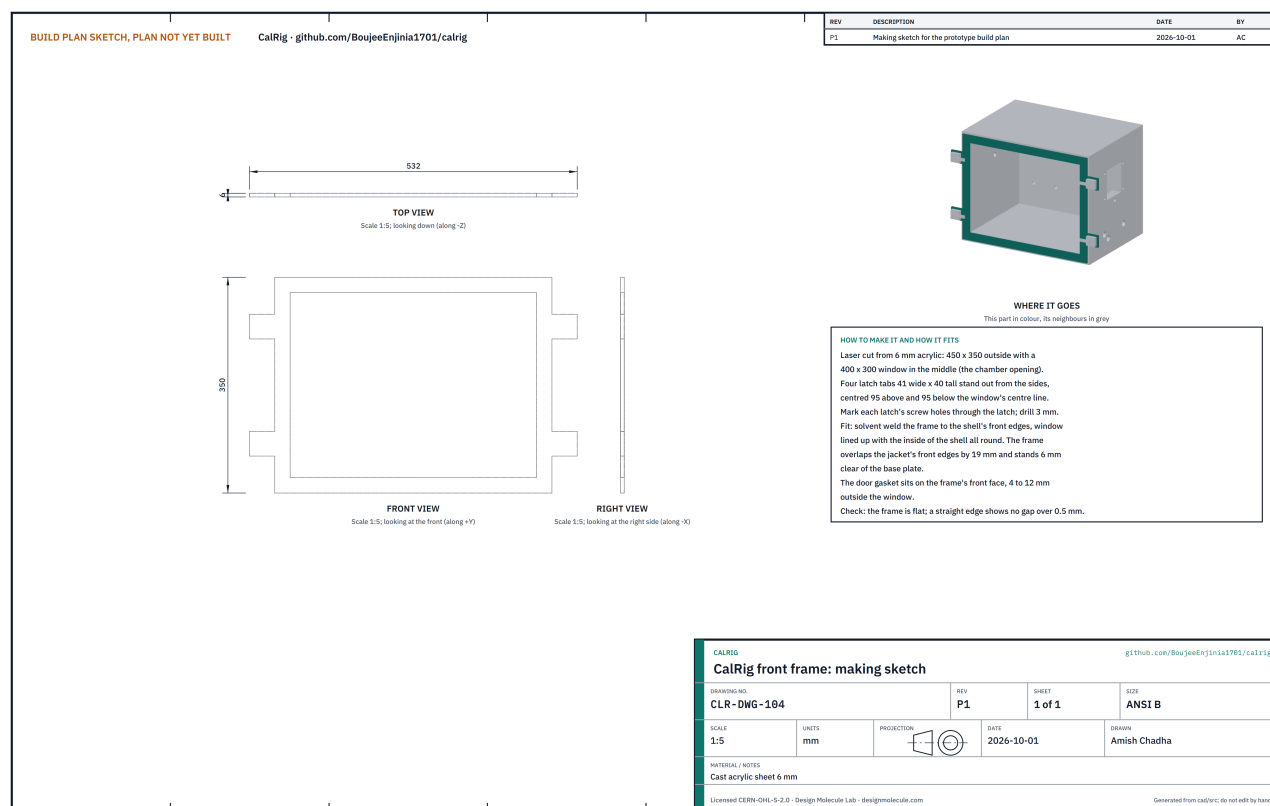


Figure 8. Front frame making sketch (CLR-DWG-104).

What it is and what it is made from. A flat frame welded to the front of the shell, which gives the door gasket a face to seal on and carries the four latches. Cast acrylic sheet 6 mm.

How to make it.

1. Laser cut a 450 x 350 mm frame with a 400 x 300 mm window in the middle.
2. Include four tabs 41 wide by 40 tall standing out from the sides, centred 95 above and 95 below the window's centre line.
3. Hold a latch on each tab, mark its screw holes and drill 3 mm.

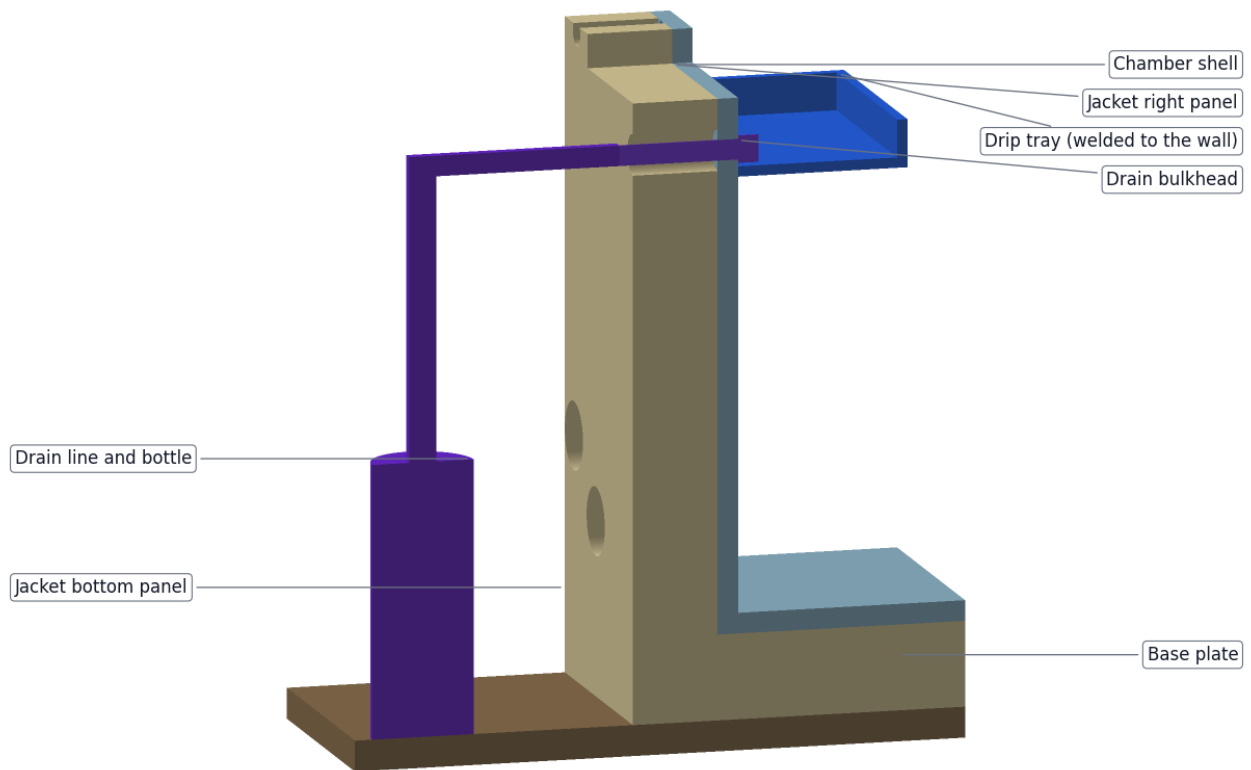
How it fits the parts next to it. Solvent weld it to the shell's front edges with the window lined up with the inside of the shell all round. It overlaps the jacket's front edges by 19 mm (Figure 5) and stands 6 mm clear of the base plate. The gasket goes on its front face, 4 to 12 mm outside the window (Figure 27).

Check before moving on. A straight edge across the frame shows no gap over 0.5 mm.

3.5 Drip tray

Joint 6: drip tray and drain (cut through the drain)

Seen from behind the cut. Water from the inner sink runs to the bottle on the base



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Figure 9. Cut through the drain: water from the inner sink collects in the tray and runs through the wall to the bottle on the base.

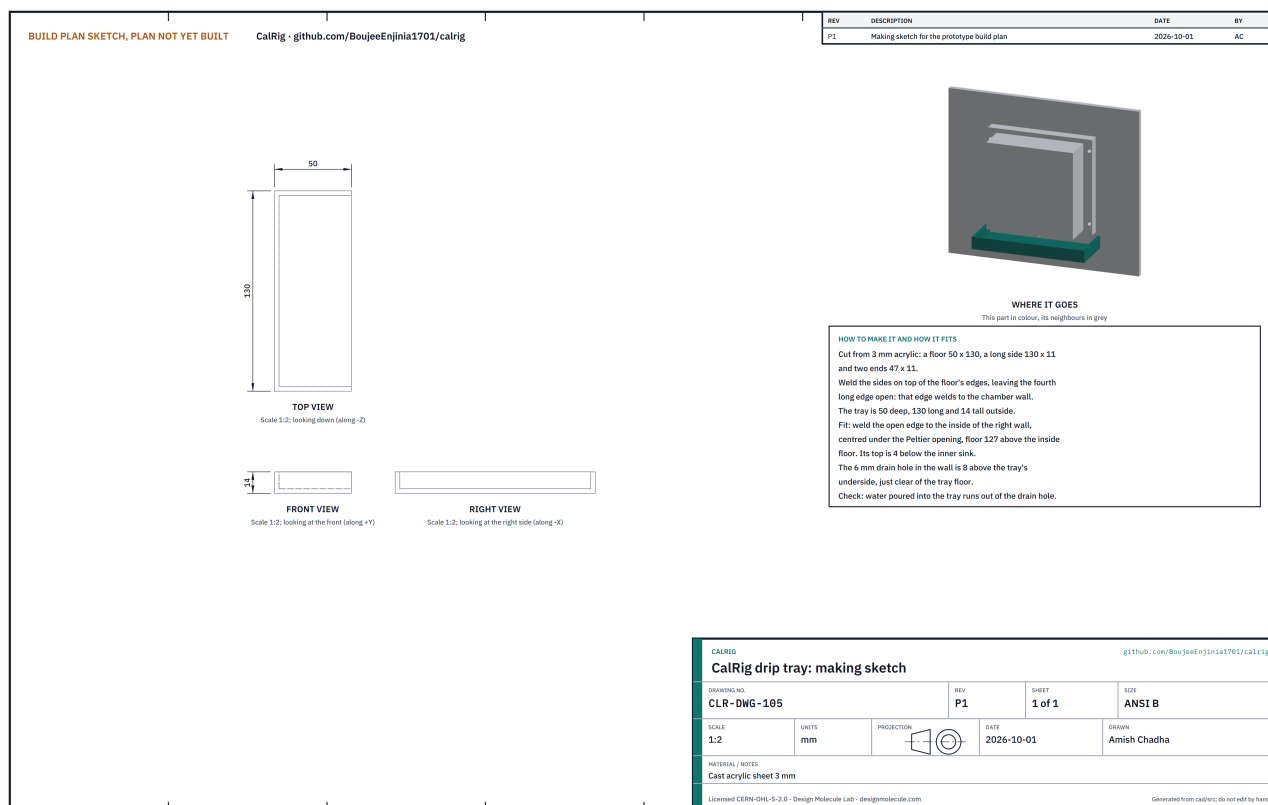


Figure 10. Drip tray making sketch (CLR-DWG-105).

What it is and what it is made from. A shallow open tray under the heat pump's inner sink that catches condensate at cold, humid set points. Cast acrylic sheet 3 mm.

How to make it.

1. Cut a floor 50 x 130 mm, one long side 130 x 11 mm and two ends 47 x 11 mm.
2. Weld the side and ends on top of the floor's edges, leaving the fourth long edge open. The tray is 50 deep, 130 long and 14 tall outside.

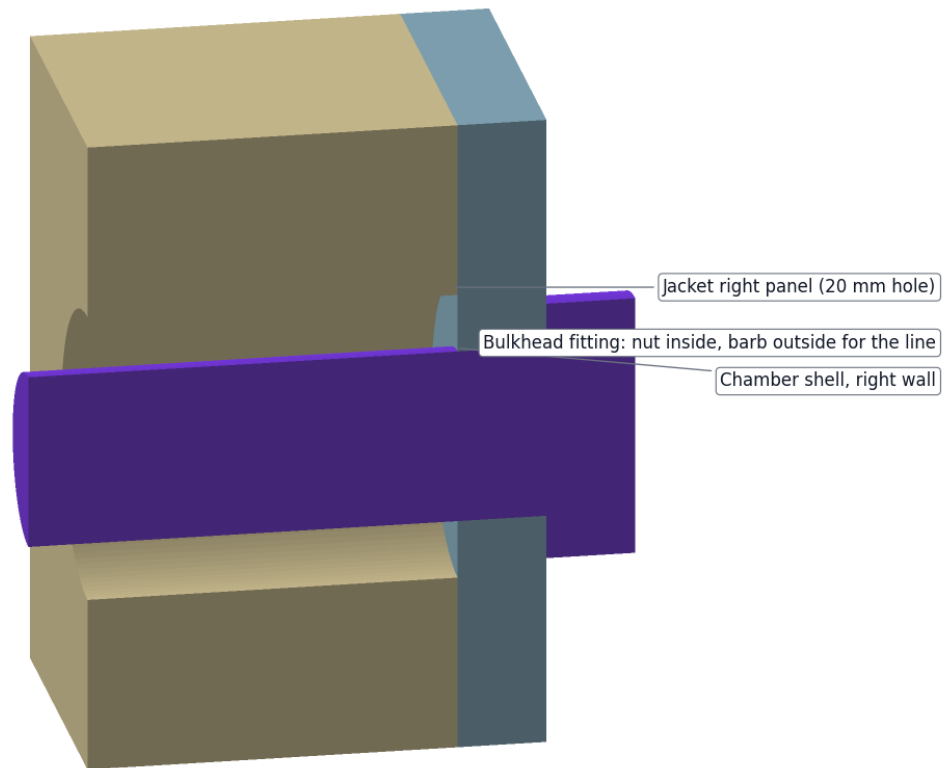
How it fits the parts next to it. Weld its open edge to the inside of the right wall, centred under the heat pump opening, with its floor 127 above the inside floor. Its top is 4 mm below the inner sink. The 6 mm drain hole in the wall is 8 mm above the tray's underside, just clear of the tray floor.

Check before moving on. Water poured into the tray runs out of the drain hole.

3.6 Bulkhead fittings, cable glands and aerosol port

Joint 2: a bulkhead fitting through the right wall (cut open)

Seen from behind the cut. The fitting seals on the acrylic; the jacket hole is 4 mm clear and filled with foam



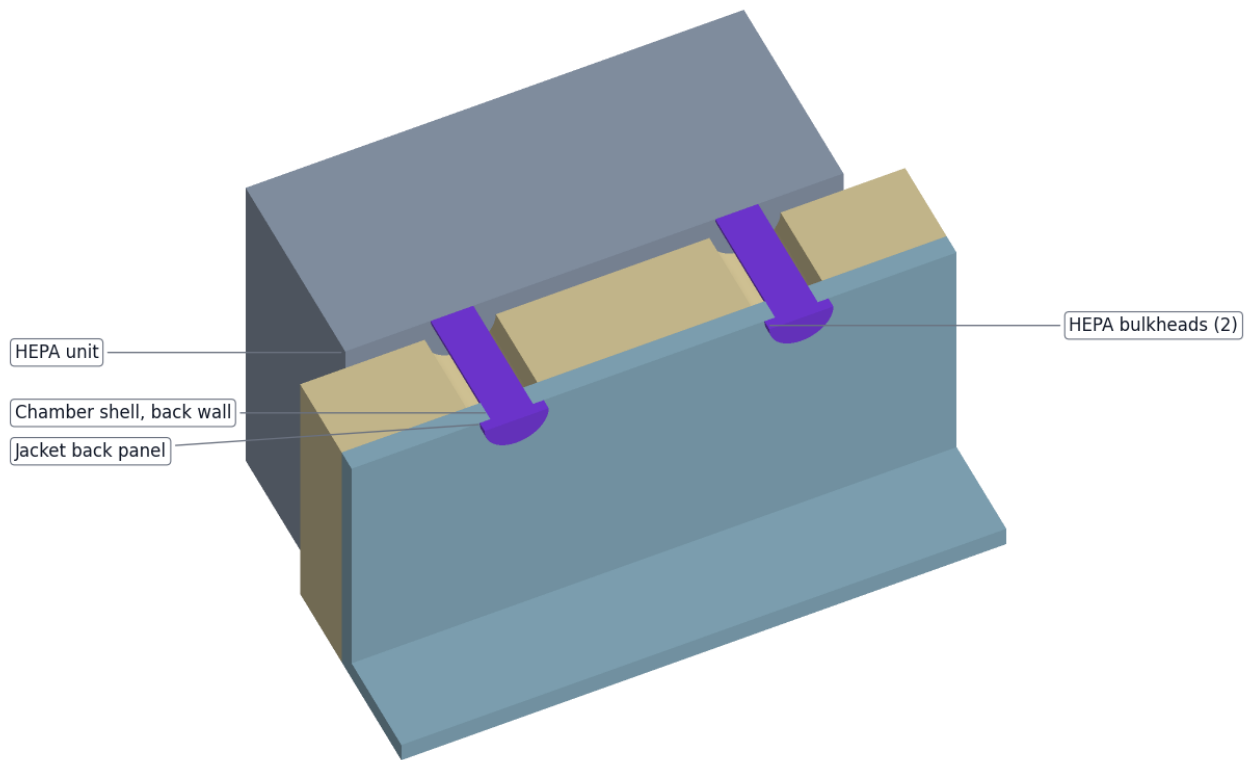
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Figure 11. A bulkhead fitting cut open: it seals on the acrylic with an O-ring outside and a nut inside; the jacket hole is 4 mm clear and is filled with foam after fitting.

Joint 7: HEPA bulkheads through the back wall (cut level with them)

Seen from above, from inside the chamber. Each fitting runs from its nut inside to the HEPA unit's port



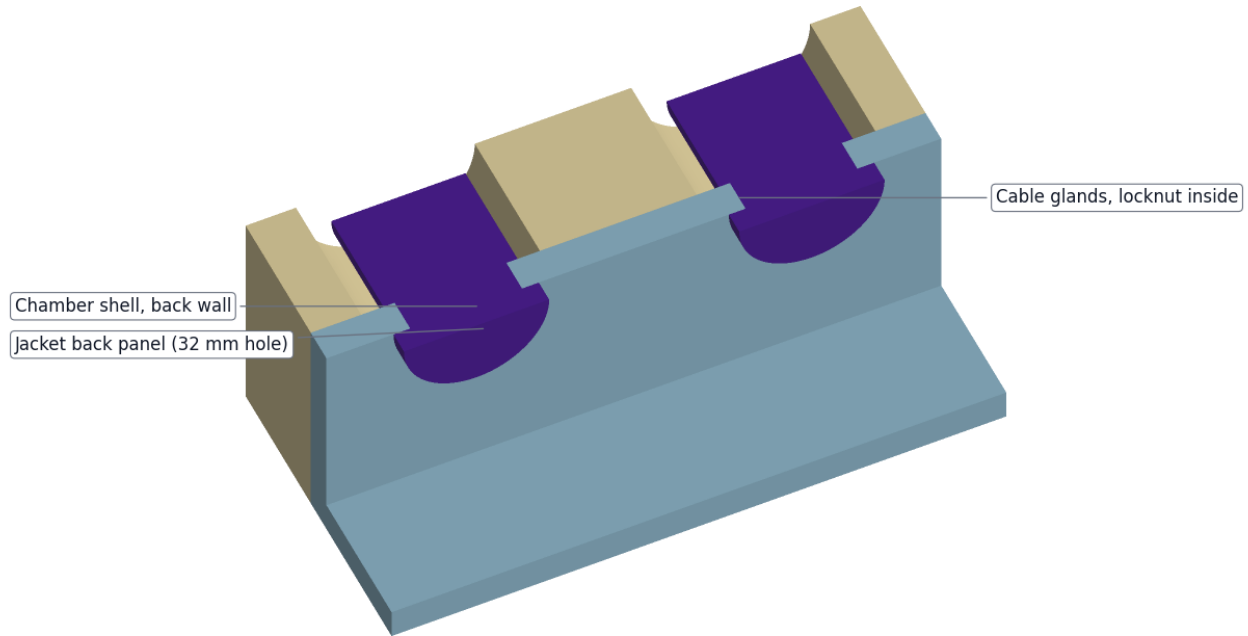
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Figure 12. The two HEPA fittings run straight from their nuts inside to the HEPA unit's ports.

Joint 8: cable glands through the back wall (cut level with them)

Seen from above, from inside the chamber. Sensor, reference and fan leads leave the chamber here



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Figure 13. The two cable glands, with their locknuts inside the chamber.

What they are. Bought fittings that take air, water and leads through the sealed walls: three 12 mm barbed bulkheads in the right wall (wet inlet at the front, dry inlet at the back, suction low between them), two 12 mm barbed bulkheads in the back wall to the HEPA unit, one 6 mm barbed bulkhead for the drain, two M20 cable glands with multi-hole seal inserts in the back wall, and the aerosol port: a 16 mm bulkhead with a Luer fitting and a small ball valve.

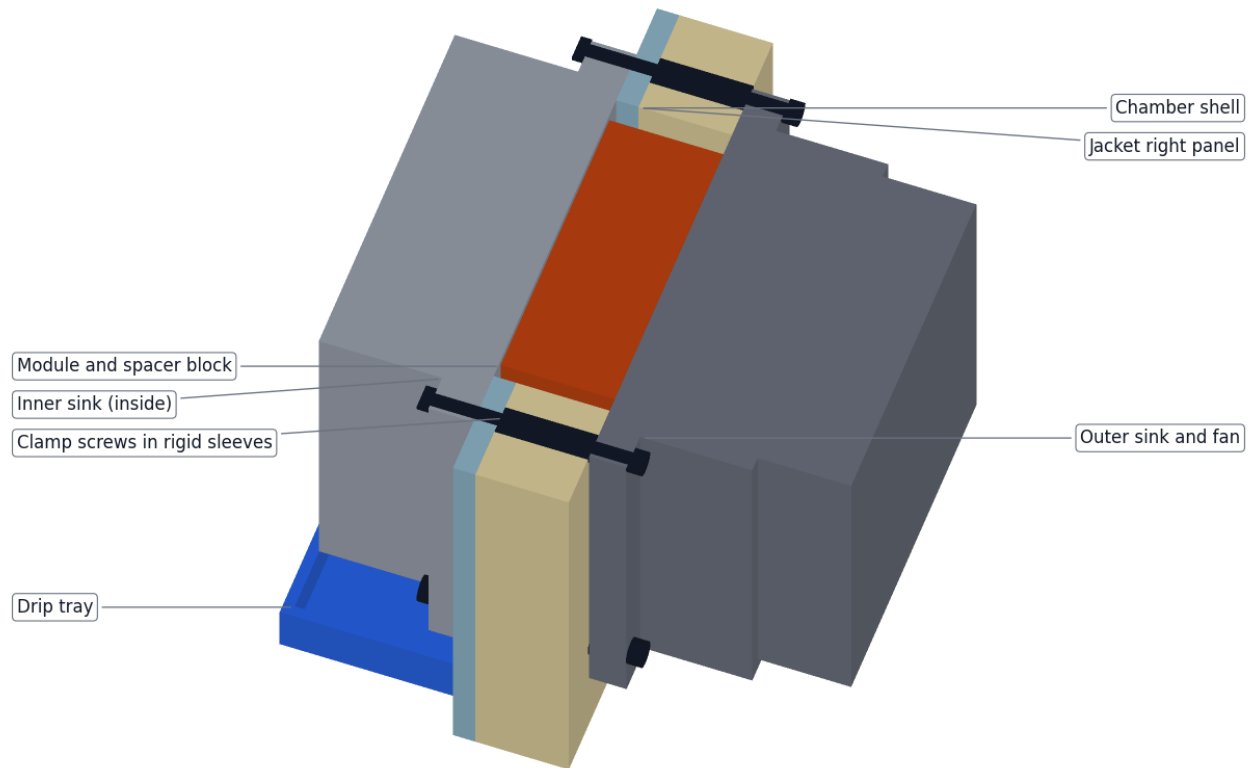
What to do to them. Nothing but fit them (step 3): each goes in from outside with its O-ring on the outside face, and its nut is tightened from inside through the open front. Check each one's thread size against the 12, 6, 20 and 16 mm holes before cutting the shell. After the jacket is on, fill the clearance round each fitting with a foam offcut and seal the surface with tape.

Check before moving on. Each fitting sits square and does not turn by hand once tight.

3.7 Peltier heat pump

Joint 3: the Peltier assembly through the right wall (cut level with the upper screws)

Seen from above. Both sink bases clamp the wall; sleeves stop the screws crushing the foam



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Figure 14. Cut level with the upper clamp screws: the inner and outer sink bases clamp the wall between them, the module and spacer block sit in the opening, and rigid sleeves carry the screws through the foam.

What it is. A bought 12 V air-to-air thermoelectric assembly of about 60 W, with its stock inner sink replaced by a larger fin block (about 45 x 120 x 110 mm) and fan. Both sink bases must be at least 120 x 110 mm.

What to do to it.

1. Drill four 4.2 mm clamp holes through each sink base, 57 mm either side of the centre and 42 mm above and below it, matching the shell's clamp holes.
2. Cut four rigid sleeves 25 mm long from 8 mm outside, 4 mm bore plastic tube.
3. Spread a thin layer of thermal paste on both faces of the module.

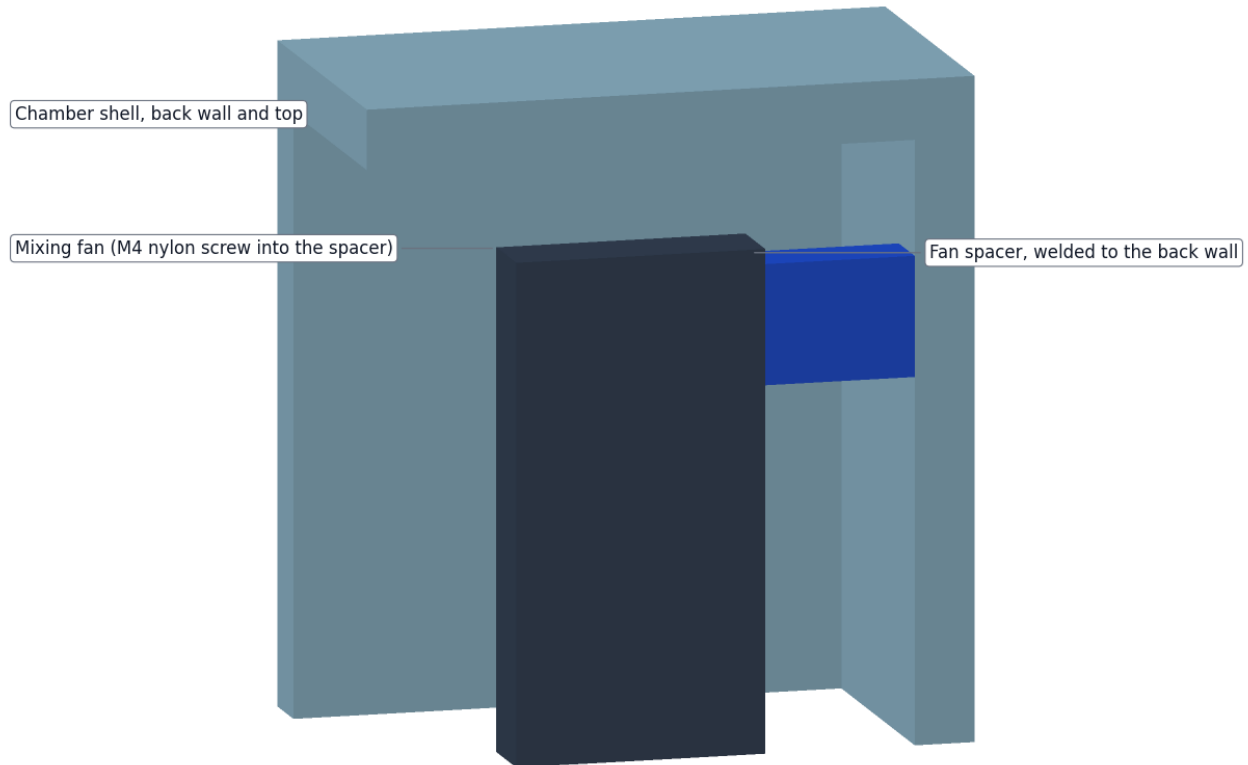
How it fits the parts next to it. The inner sink's base sits flat on the inside of the right wall over the opening, with a thin foam gasket; the outer sink's base sits flat on the jacket over the same opening. The 80 x 80 mm module and spacer block sits between them in the 90 x 100 mm opening, 5 mm clear all round; fill that gap with foam offcuts and sealant. Four M4 stainless screws pass through both bases, the wall and the sleeves; tighten them evenly in a cross pattern so the module is squeezed flat.

Check before moving on. Both sinks sit flat; the module cannot be moved; nothing touches the acrylic but the inner sink's base and its gasket.

3.8 Mixing fan

Joint 4: mixing fan on its spacers (cut through the upper left spacer)

Seen from the right, inside the chamber. The fan stands 15 mm off the back wall so it can draw air



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Figure 15. Cut through the upper left spacer: the fan stands 15 mm off the back wall on its welded spacers.

What it is. A bought 120 mm, 12 V fan with speed control and a finger guard.

How it fits the parts next to it. It screws to the four spacers on the back wall with four M4 nylon screws, blowing toward the door, 10 mm below the top and 20 mm from the left wall. Its lead runs down the back wall to a cable gland.

Check before moving on. The blades turn freely by hand and clear the guard.

3.9 Sensor tray

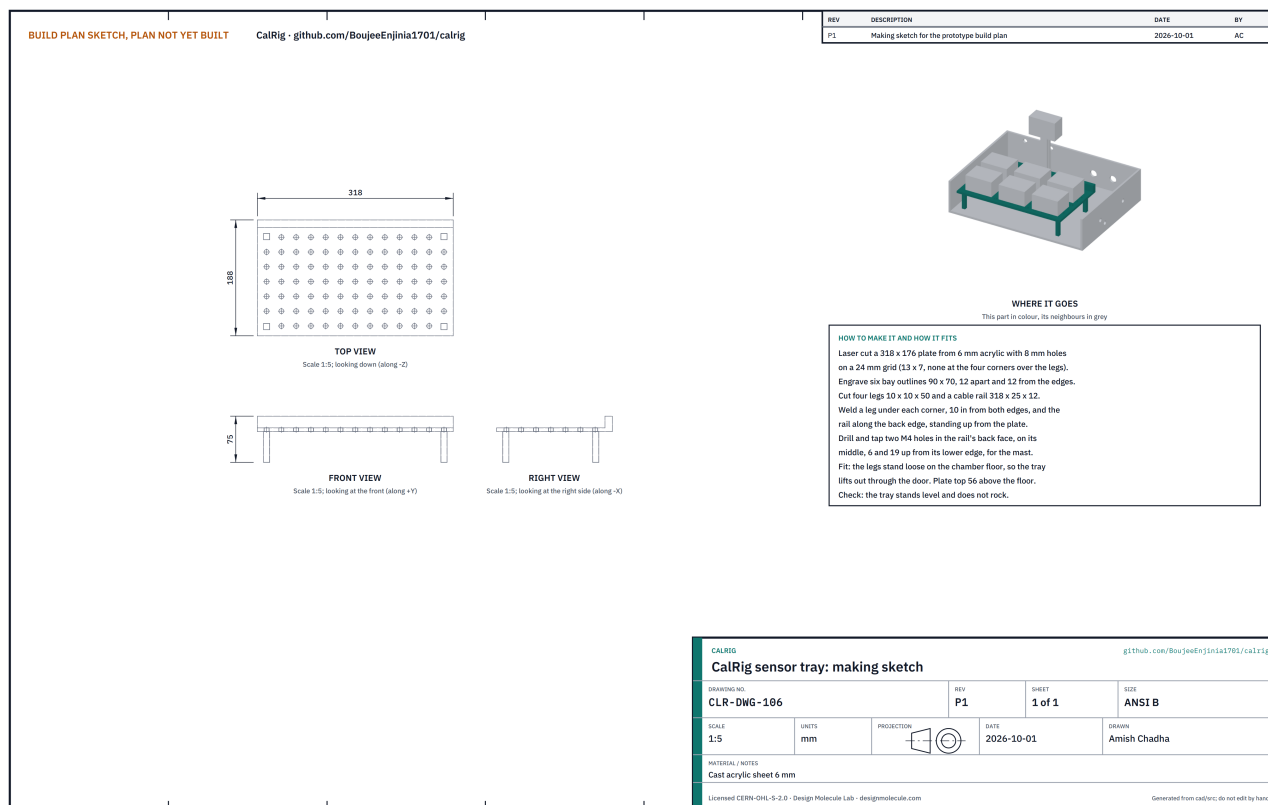


Figure 16. Sensor tray making sketch (CLR-DWG-106).

What it is and what it is made from. The perforated shelf that holds six sensor heads at the largest size the rig takes (90 x 70 x 50 mm), with a cable rail along its back. Cast acrylic sheet 6 mm.

How to make it.

1. Laser cut a 318 x 176 mm plate with 8 mm holes on a 24 mm grid (13 by 7), leaving out the four corner holes. Engrave six bay outlines 90 x 70 mm, 12 mm apart and 12 mm from the edges.
2. Cut four legs 10 x 10 x 50 mm and a rail 318 x 25 x 12 mm.
3. Weld a leg under each corner, 10 mm in from both edges, and the rail along the back edge, standing up from the plate.
4. Drill and tap two M4 holes in the rail's back face, on its middle, 6 and 19 mm up from its lower edge.

How it fits the parts next to it. The legs stand loose on the chamber floor, 10 mm from the left wall and 16 mm behind the front opening; the plate top is 56 mm above the floor and the tray lifts out through the door. Nothing on it touches the drip tray or the inner sink.

Check before moving on. The tray stands level and does not rock.

3.10 Reference mast and cluster

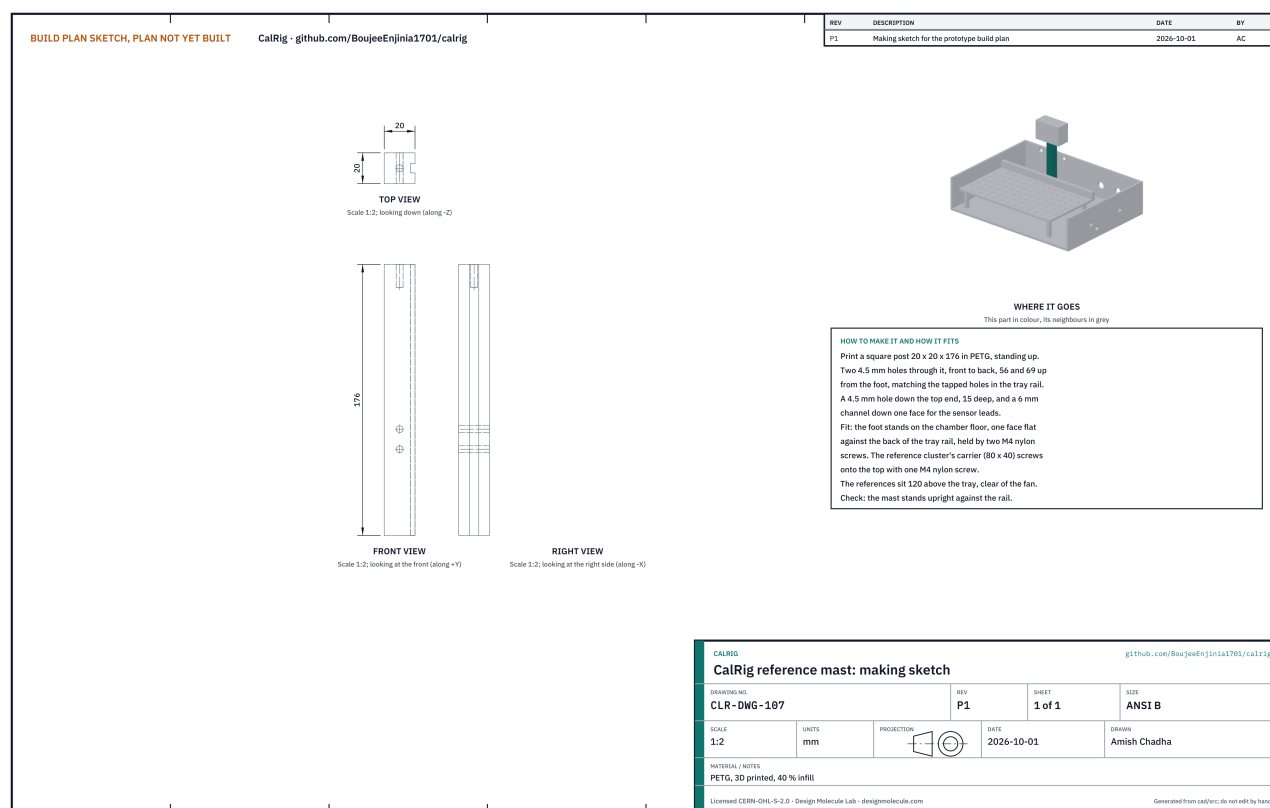
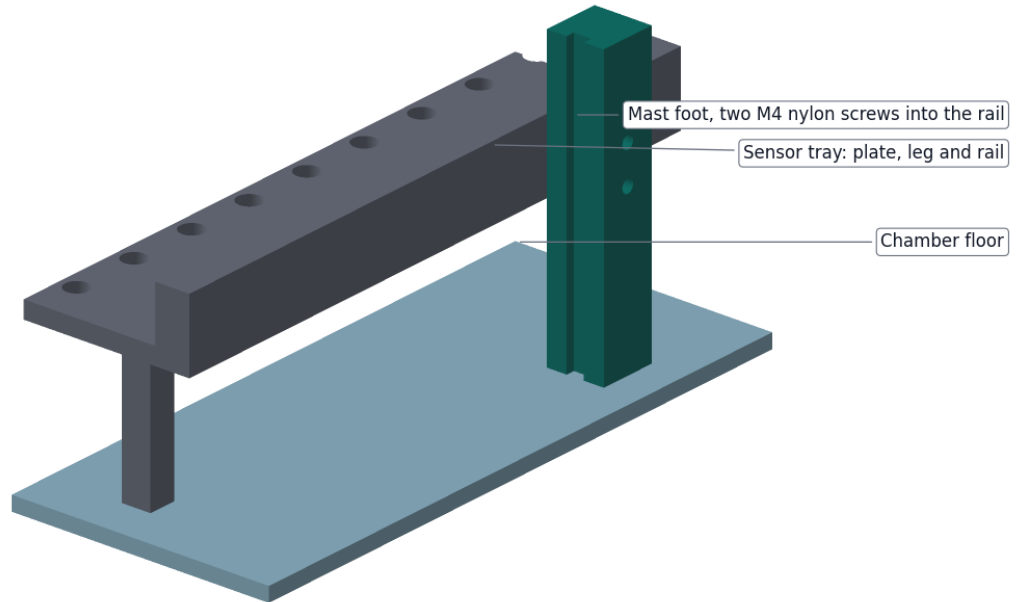


Figure 17. Reference mast making sketch (CLR-DWG-107).

Joint 5: tray rail and mast foot (behind the tray)

Seen from behind and to the right. Mast and legs stand on the floor; nothing is screwed into the shell



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Figure 18. The mast stands on the floor against the back of the tray rail; the tray's legs stand on the floor too, so nothing is screwed into the shell.

What it is and what it is made from. A square post that holds the reference cluster (two SHT45 temperature and humidity sensors, an SCD30 carbon dioxide sensor and the SPS30 particle sensor on an 80 x 40 mm carrier) 120 mm above the tray. PETG, printed standing up, 40 % infill.

How to make it.

1. Print a 20 x 20 x 176 mm post with two 4.5 mm holes through it front to back, 56 and 69 mm up from the foot, a 4.5 mm hole 15 mm deep down its top end, and a 6 mm channel down one side for the leads.
2. Mount the four reference sensors on their carrier as their makers describe.

How it fits the parts next to it. The foot stands on the chamber floor with one face flat against the back of the tray rail, held by two M4 nylon screws into the rail. The carrier screws onto the top with one M4 nylon screw. The leads run down the channel to a cable gland.

Check before moving on. The mast stands upright; the references clear the mixing fan by 30 mm.

3.11 Dryer socket

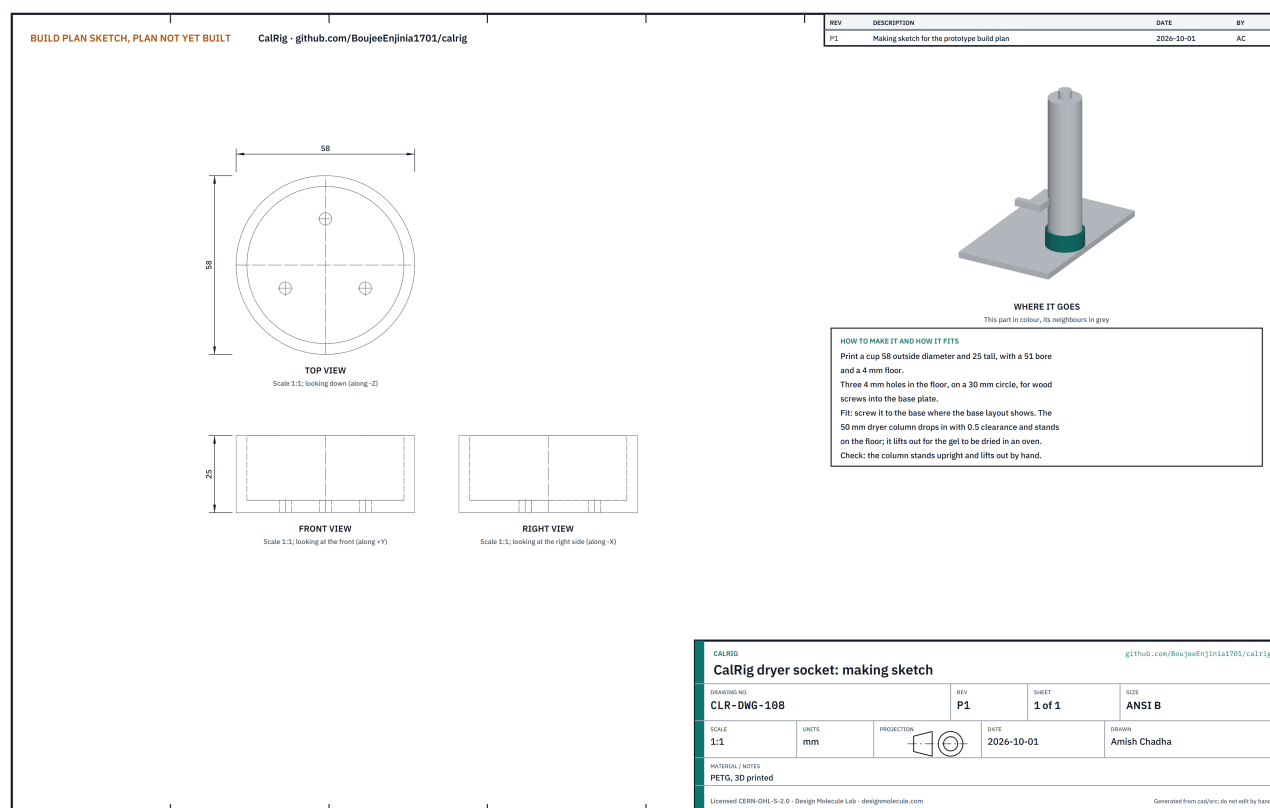


Figure 19. Dryer socket making sketch (CLR-DWG-108).

What it is and what it is made from. A cup that holds the 50 mm dryer column upright on the base. PETG, printed.

How to make it. Print a cup 58 mm outside diameter and 25 mm tall with a 51 mm bore, a 4 mm floor and three 4 mm screw holes on a 30 mm circle.

How it fits the parts next to it. Screw it to the base where Figure 3 shows. The column drops in with 0.5 mm clearance and lifts out so the gel can be dried in an oven.

Check before moving on. The column stands upright and lifts out by hand.

3.12 Jar rack

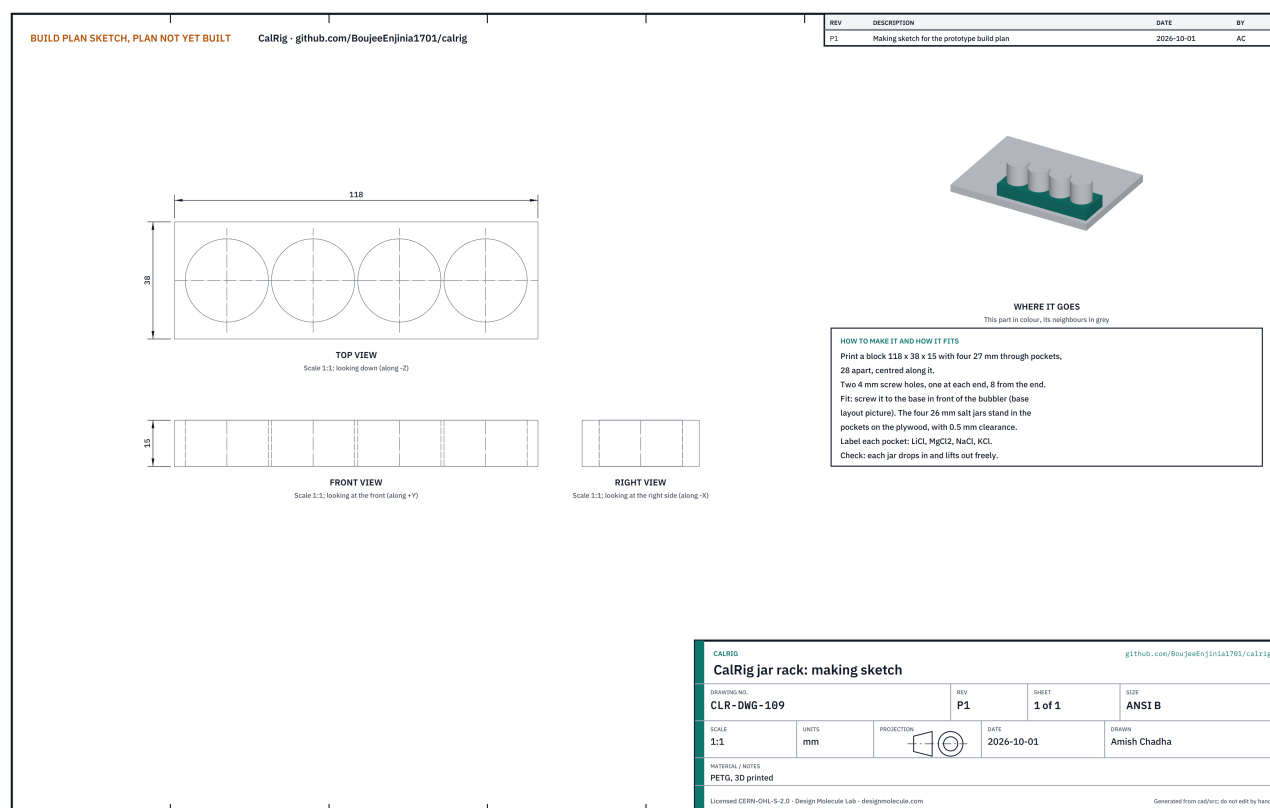


Figure 20. Jar rack making sketch (CLR-DWG-109).

What it is and what it is made from. A block that keeps the four salt jars upright and in order. PETG, printed.

How to make it. Print a block 118 x 38 x 15 mm with four 27 mm pockets 28 mm apart and a 4 mm screw hole 8 mm from each end. Label the pockets LiCl, MgCl₂, NaCl and KCl.

How it fits the parts next to it. Screw it to the base in front of the bubbler (Figure 3). The 26 mm jars stand on the plywood in the pockets with 0.5 mm clearance.

Check before moving on. Each jar drops in and lifts out freely.

3.13 Door

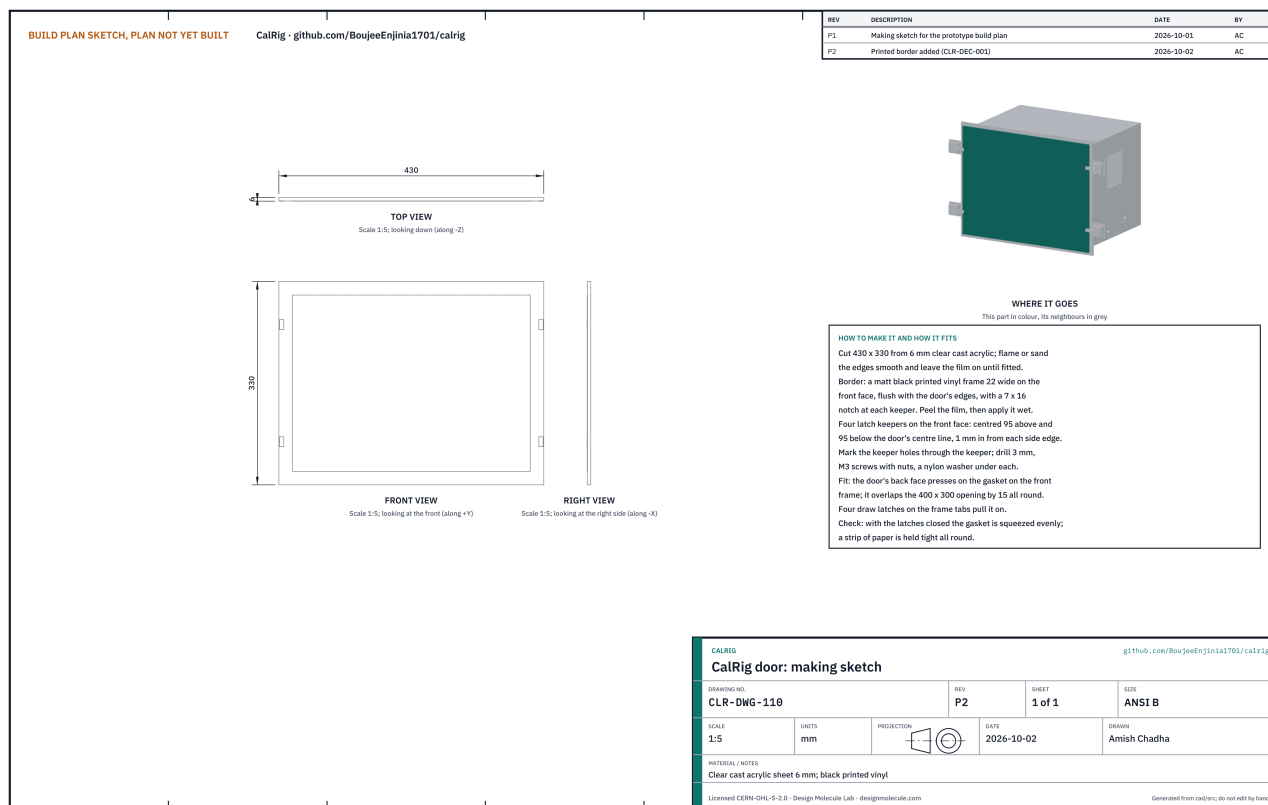


Figure 21. Door making sketch (CLR-DWG-110).

What it is and what it is made from. The clear front of the chamber, held against the gasket by four draw latches, with a printed black border round its edge. Clear cast acrylic sheet 6 mm and printed vinyl.

How to make it.

1. Cut 430 x 330 mm. Sand or flame the edges smooth and leave the protective film on until it is fitted.
2. Peel the film from the front face and apply the border: a matt black printed vinyl frame 22 mm wide, flush with the door's edges, with a 7 x 16 mm notch at each keeper position. Apply it wet and squeeze out the bubbles.
3. Place a latch keeper on the front face at each latch: centred 95 mm above and 95 mm below the door's centre line, 1 mm in from each side edge. Mark the holes through the keeper, drill 3 mm, and fit M3 screws with nuts and a nylon washer under each.

How it fits the parts next to it. The back face presses on the gasket on the front frame and overlaps the 400 x 300 mm opening by 15 mm all round. The latches on the frame tabs hook the keepers (Figure 27).

Check before moving on. The border lies flat with no bubbles at the edges. With the latches closed, a strip of paper trapped between door and gasket is held tight all round.

3.14 Door panel

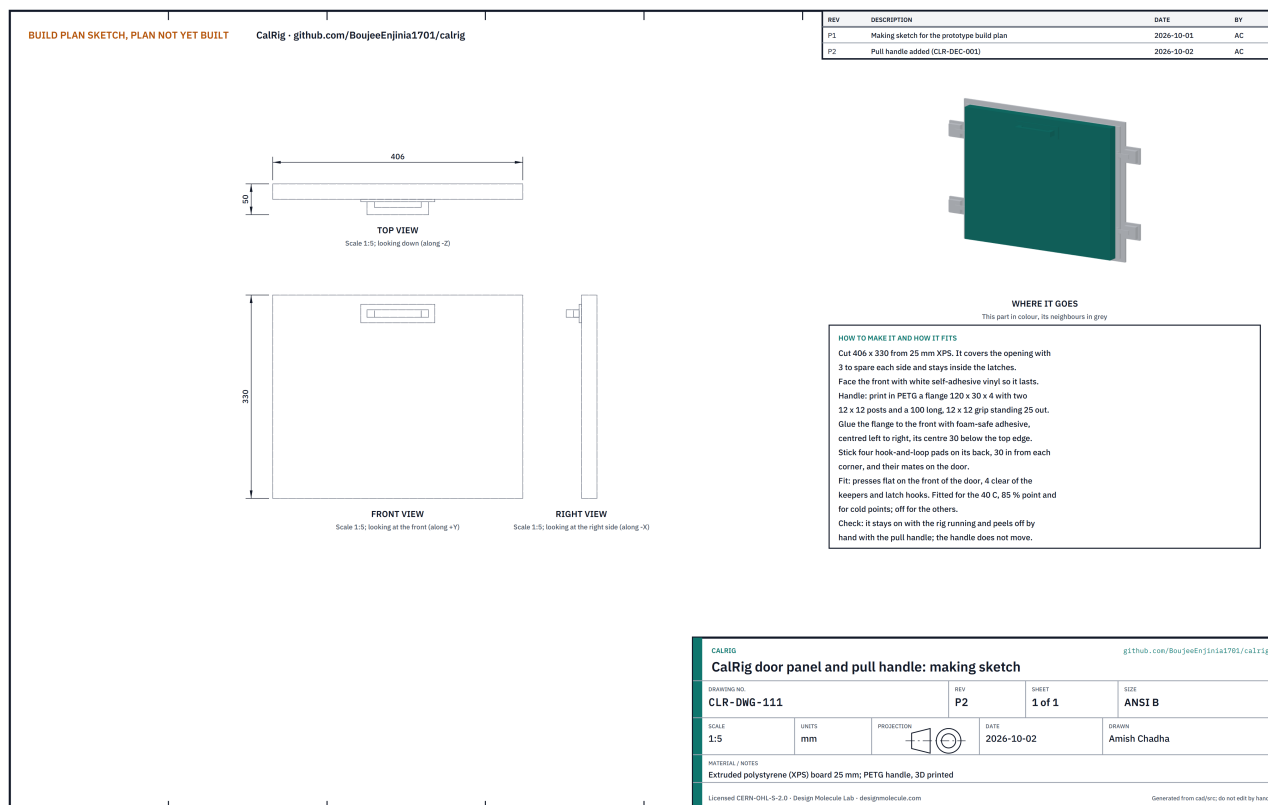


Figure 22. Door panel and pull handle making sketch (CLR-DWG-111).

What it is and what it is made from. A removable foam panel that insulates the door for the hot, humid set point and for cold points, with a pull handle on its front. Extruded polystyrene board 25 mm; the handle is PETG, printed.

How to make it.

1. Cut 406 x 330 mm, face the front with white self-adhesive vinyl, and stick four hook-and-loop pads on its back, 30 mm in from each corner, with their mates on the door.
2. Print the pull handle: a flange 120 x 30 x 4 mm, two 12 x 12 mm posts and a grip 100 mm long and 12 x 12 mm in section, standing 25 mm out from the flange.
3. Glue the flange to the front with foam-safe adhesive, centred left to right, its centre 30 mm below the top edge. Leave it a day before pulling on it.

How it fits the parts next to it. It presses flat on the front of the door between the latches, 4 mm clear of the keepers and latch hooks, and covers the opening with 3 mm to spare each side.

Check before moving on. It stays on with the fans running and peels off by the handle without the handle moving.

3.15 Front badge and status light

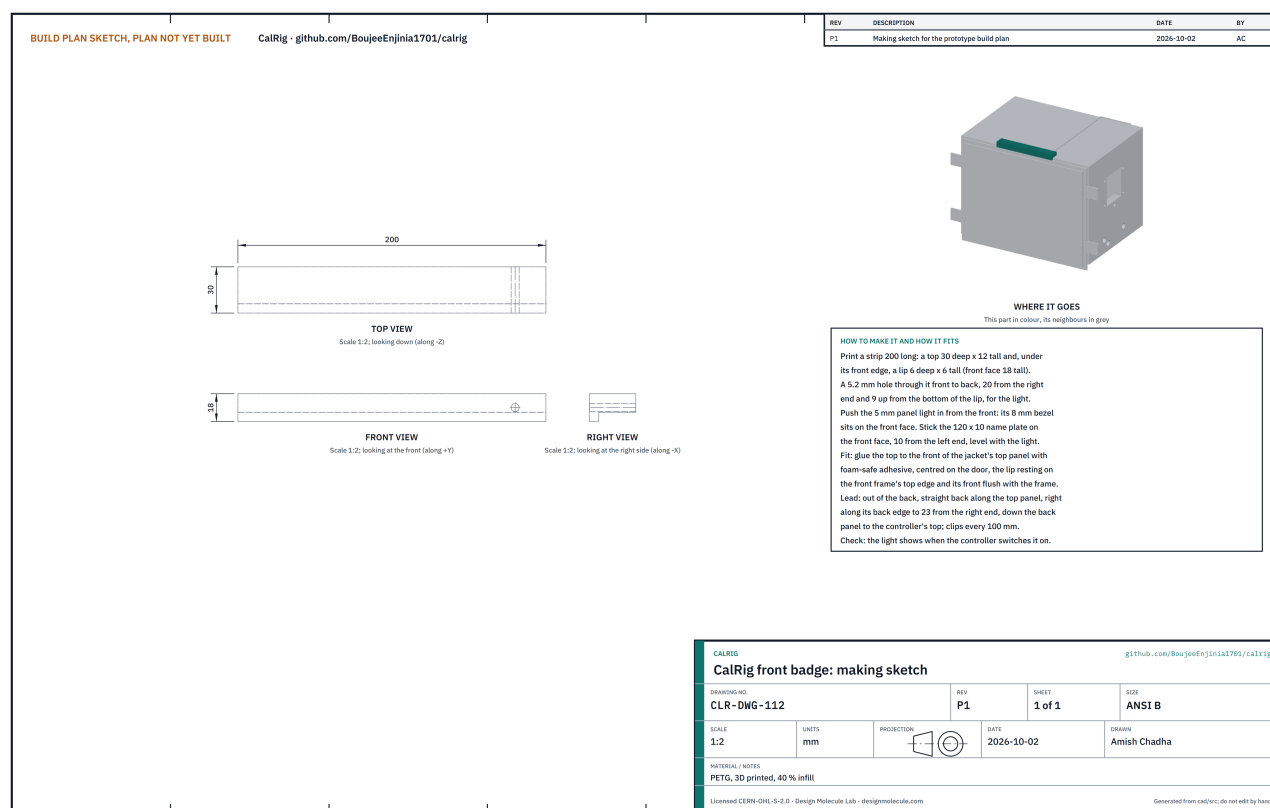
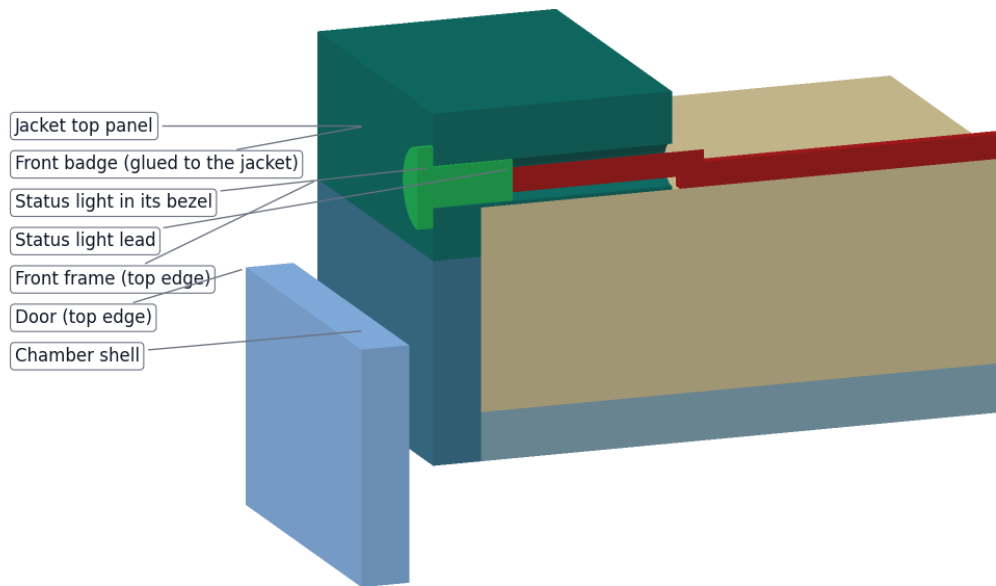


Figure 23. Front badge making sketch (CLR-DWG-112).

Joint 10: front badge and status light (cut through the light)

Seen from the right, slightly in front, on the cut. The badge sits on the jacket's top panel, its lip on the frame's top edge



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Figure 24. Cut through the light: the badge sits on the jacket's top panel with its lip on the front frame's top edge; the light's lead leaves through the back of the badge.

What it is and what it is made from. A strip on the front of the rig that carries the name plate and a status light, so a run can be seen from across the room. PETG, printed, 40 % infill; a bought 12 V panel light (5 mm, in an 8 mm bezel with its resistor built in); a 120 x 10 mm name plate label; a two-core 0.25 mm² lead about 0.9 m long with a plug for the controller.

How to make it.

1. Print a strip 200 mm long: a top 30 mm deep and 12 mm tall and, under its front edge, a lip 6 mm deep and 6 mm tall, so the front face is 18 mm tall.
2. Include a 5.2 mm hole through it from front to back, 20 mm from the right-hand end and 9 mm up from the bottom of the lip.
3. Push the panel light in from the front so its bezel sits on the front face, and feed its lead out of the back.
4. Stick the name plate on the front face, 10 mm from the left-hand end, level with the light.

How it fits the parts next to it. Glue the top to the front of the jacket's top panel with foam-safe adhesive, centred on the door, with the lip resting on the front frame's top edge and its front flush with the frame (Figure 24). Run the lead straight back along the top panel, then right along the panel's back edge to 23 mm from its right-hand end, then down the back panel to the top of the controller, with a stick-on clip every 100 mm. Plug it into the controller's status light output.

Check before moving on. The badge does not touch the door or the door panel; the light comes on when the controller switches it.

3.16 Air loops and wiring

CalRig prototype: air loops and drain

Every loop starts and ends in the chamber, so the chamber stays sealed. Tube is 6 mm bore silicone, clamped on every barb.

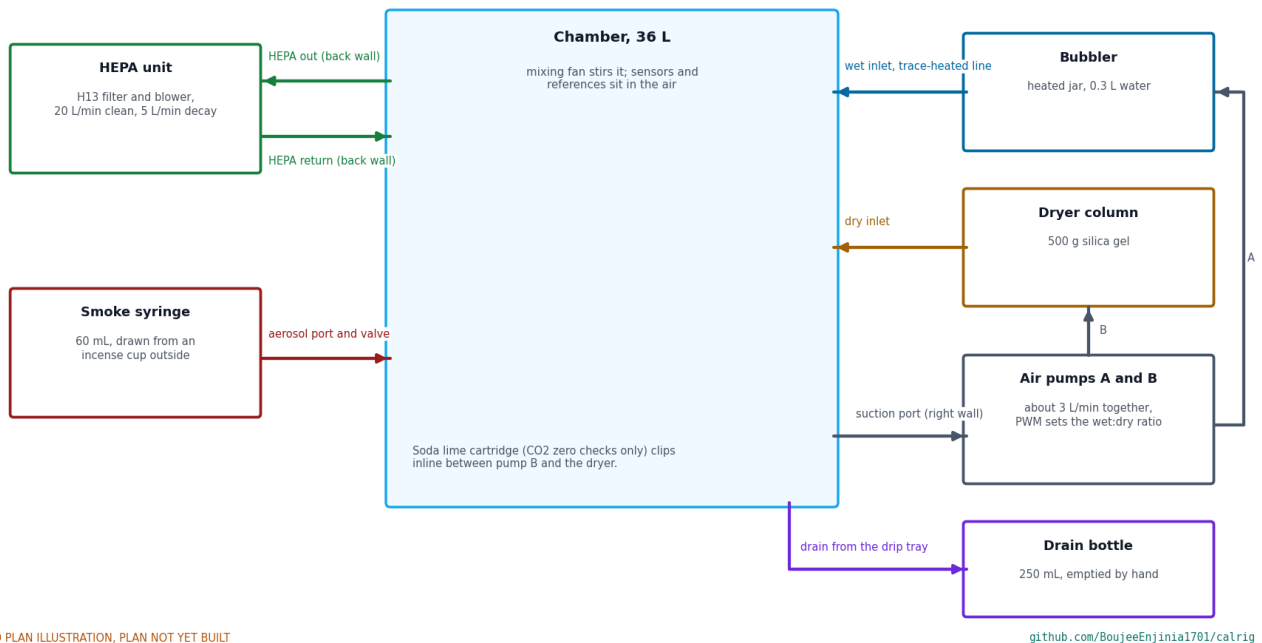
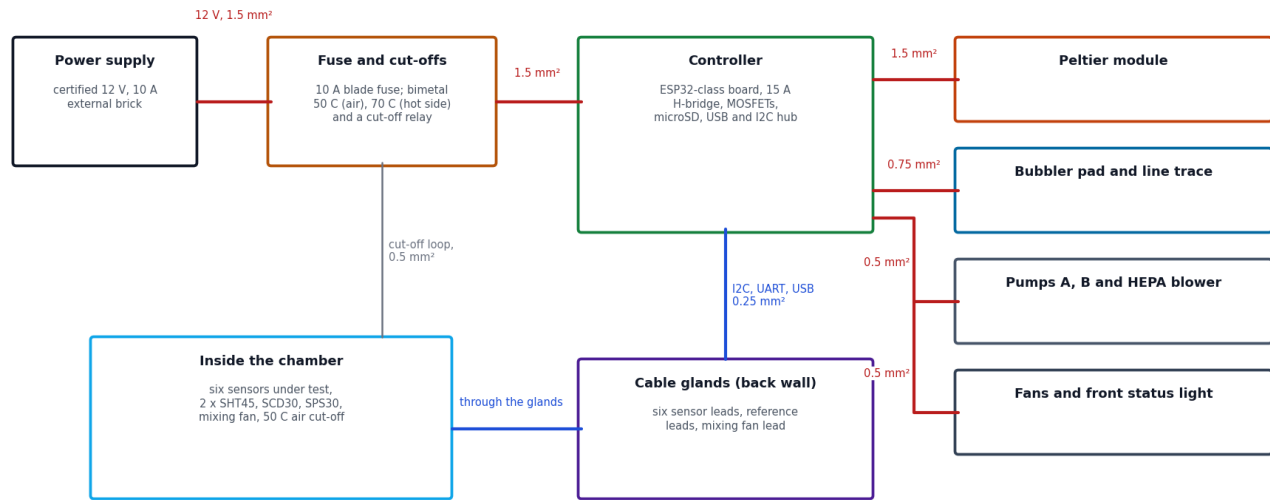


Figure 25. Every air loop starts and ends in the chamber, so the chamber stays sealed.

CalRig prototype: block-level wiring

Bought modules wired at block level; no circuit board is laid out. Stranded copper; ferrules on every screw terminal.



Safety: the two bimetal cut-offs hold in the relay that feeds the Peltier and heater outputs; if either opens, those outputs lose power whatever the software does.

Red: 12 V power. Blue: data. Grey: cut-off loop and thermistors (bubbler pad, line trace and outer sink each carry one). Everything is 12 V; no mains inside the rig.

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Figure 26. Block-level wiring with wire sizes. No circuit board is laid out at this stage; bought modules stand in for the power board.

The power board in the bill of materials is a set of bought modules for this prototype:

Table 2. Modules that make up the controller and power board.

Module	What to buy
Controller	ESP32-class development board with a microSD socket and a USB and I2C hub for the six sensors and the references
Heat pump driver	12 V H-bridge rated 15 A or more, so the controller can heat or cool
Fan, pump and heater switches	Logic-level MOSFET modules, one per load, with flyback diodes on the fans and pumps; one more output for the front status light
Cut-offs	Two normally closed bimetal switches, 50 °C (stuck to the inside of the top, near the fan) and 70 °C (screwed to the outer sink's base), in series with the coil of a 12 V relay rated 10 A that feeds the heat pump and heater outputs
Fuse	10 A blade fuse in an inline holder at the supply input

Wire it like this, with stranded copper and a ferrule on every screw terminal:

1. Supply to the fuse, and the fuse to the controller's power input: 1.5 mm² (16 AWG).
2. Heat pump driver to the module: 1.5 mm².
3. Heater switch to the bubbler pad and line trace: 0.75 mm² (18 AWG).
4. Switches to the pumps, HEPA blower and fans: 0.5 mm² (20 AWG).

5. The two cut-offs in series with the relay coil: 0.5 mm², twisted.
6. Sensors and references through the cable glands to the hub: 0.25 mm² (24 AWG) or their own leads.
7. Thermistors on the bubbler pad, the line trace and the outer sink to the controller: 0.25 mm², twisted.
8. The front status light's lead to the controller's status light output: 0.25 mm², two cores, plugged in.

Run the tubes as Figure 25 shows, with 6 mm bore silicone tube and a clamp on every barb. The soda lime cartridge clips in between pump B and the dryer only for carbon dioxide zero checks.

Check before moving on. Every wire continues end to end; with the supply off, the 12 V bus reads open to ground; every wire and tube is labelled.

3.17 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Peltier heat pump (line 5).** As section 3.7; sink bases at least 120 x 110 mm.
- **Mixing fan (line 6).** 120 mm, 12 V, about 85 m³/h free air, speed control, guard.
- **Reference cluster (line 8).** Two SHT45 breakouts, an SCD30 and an SPS30 that has been collocated at a monitoring station before use.
- **Bubbler (line 9).** 500 mL glass jar filled to 300 mL, air stone, 10 W 12 V heater pad with thermistor, 10 mm foam sleeve, 12 V diaphragm pump of about 1.5 L/min, outlet line with a 3 W trace heater under foam.
- **Dryer column (line 10).** 50 mm acrylic tube about 230 mm long with 500 g of indicating silica gel, and the second pump.
- **HEPA unit (line 11).** H13 filter cartridge and a 60 mm variable-speed blower in a housing with two 12 mm ports 80 mm apart, or one you can drill to match.
- **Aerosol port (line 12).** 16 mm bulkhead with a female Luer fitting, small ball valve, 60 mL syringe, metal incense cup.
- **Controller and power board (line 13).** As Table 2.
- **Power supply (line 14).** Certified external 12 V, 10 A supply with a regional mains lead.
- **Salt jars (line 15).** Four sealed jars of saturated LiCl, MgCl₂, NaCl and KCl slurry with sensor lid adapters.
- **Door hardware (line 2).** Silicone D-section gasket about 10 mm wide; four small draw latches with keepers that fit a 41 x 40 mm tab; a matt black vinyl border 430 x 330 mm outside and 22 mm wide, cut by a sign or print shop.
- **Front badge parts (line 20).** A 12 V panel indicator light, 5 mm, in an 8 mm bezel with a built-in resistor; a name plate label 120 x 10 mm; two-core 0.25 mm² lead with a plug; stick-on cable clips.
- **Fittings (line 19).** As section 3.6, plus a 250 mL bottle with a lid and 6 mm drain tube.
- **Wiring and consumables (line 17).** Wire, ferrules, silicone tube and clamps, foam-safe adhesive, foil tape, acrylic solvent cement, M3 and M4 nylon and stainless screws, wood screws, hook-and-loop pads, labels.

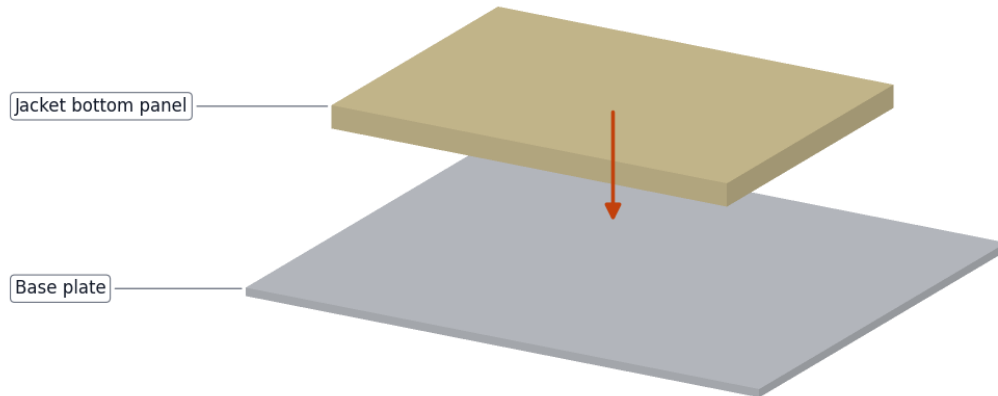
■ 4. Putting it together

In each picture the parts already fitted are grey and the parts being fitted are in colour, with an arrow showing the way they go in.

Step 1: jacket bottom panel onto the base plate

Step 1: jacket bottom panel onto the base plate

Foam-safe adhesive; left edge 46 mm and front edge 94 mm in from the base edges (base layout)



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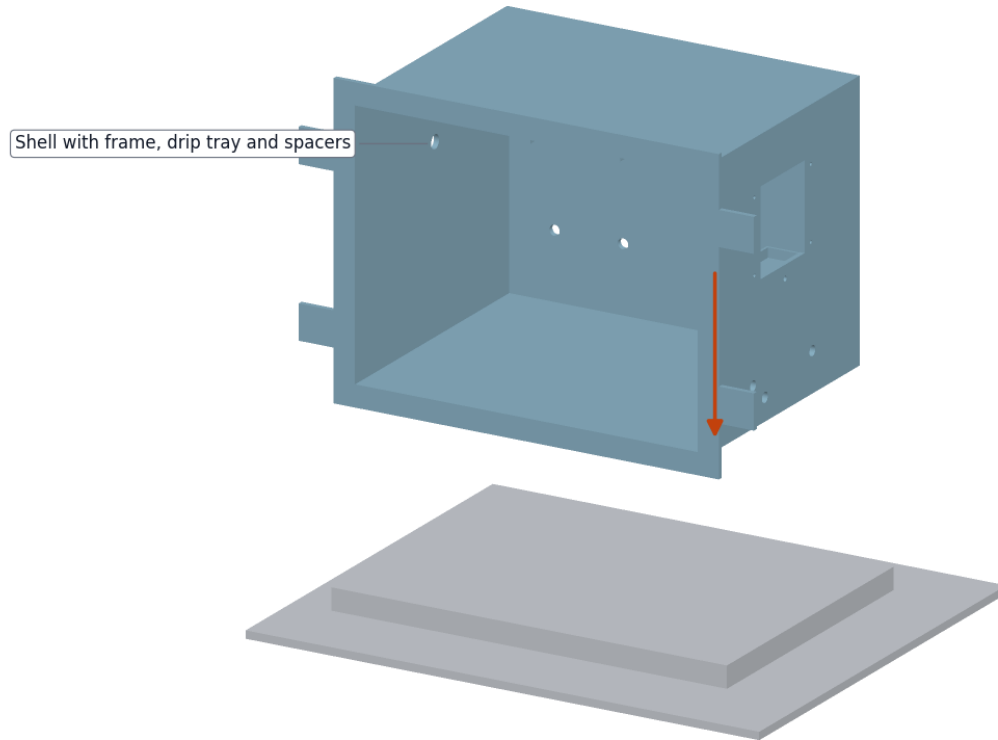
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Foam-safe adhesive on the base; the panel 46 mm from the left edge and 94 mm from the front edge. Weigh it down until the adhesive sets.

Step 2: chamber shell onto the jacket

Step 2: chamber shell onto the jacket

Lower it on, frame forward; foam-safe adhesive under the shell's bottom



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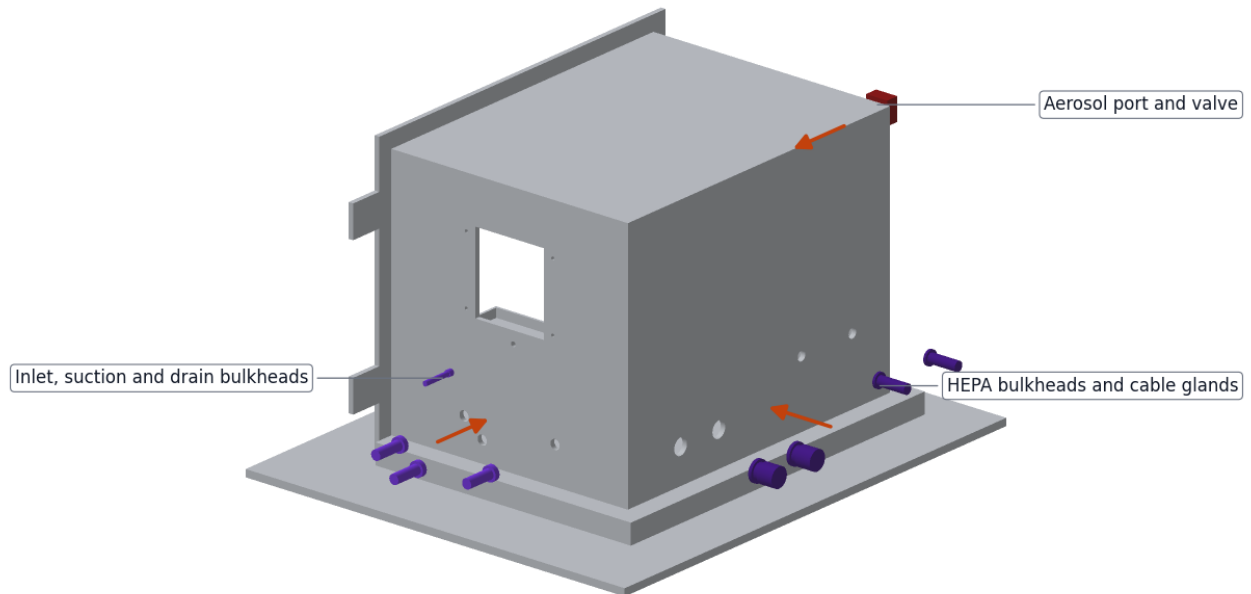
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The shell, with its frame, drip tray and spacers already welded, goes down onto the bottom panel, frame forward, on a bead of foam-safe adhesive. Its sides line up with the panel's sides.

Step 3: bulkheads, glands and aerosol port into the walls

Step 3: bulkheads, glands and aerosol port into the walls

Each from outside with its O-ring; locknut inside, reached through the open front. Seen from behind right



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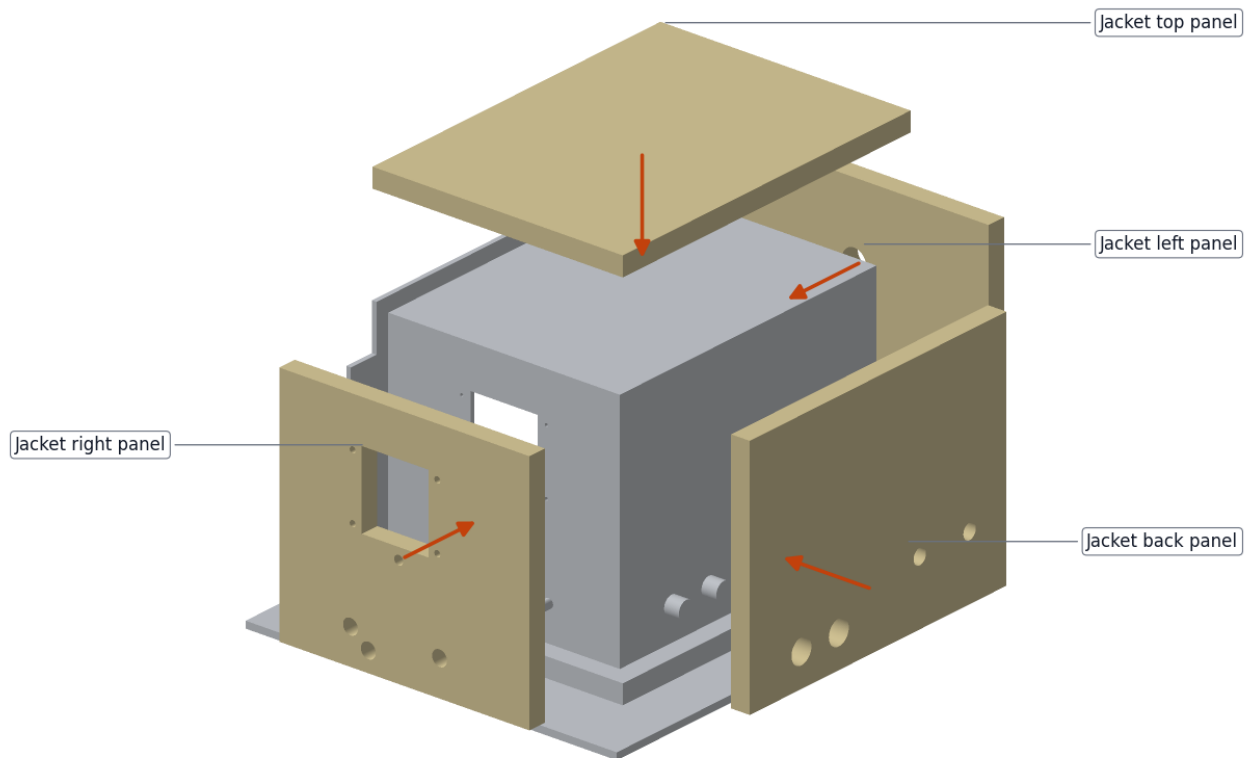
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Each from outside with its O-ring on the outside face; nut or locknut tightened from inside through the open front. Seen from behind and to the right.

Step 4: jacket back, side and top panels

Step 4: jacket back, side and top panels

Slide each over the fittings; foam-safe adhesive on the shell; foil tape on every seam. Seen from behind right



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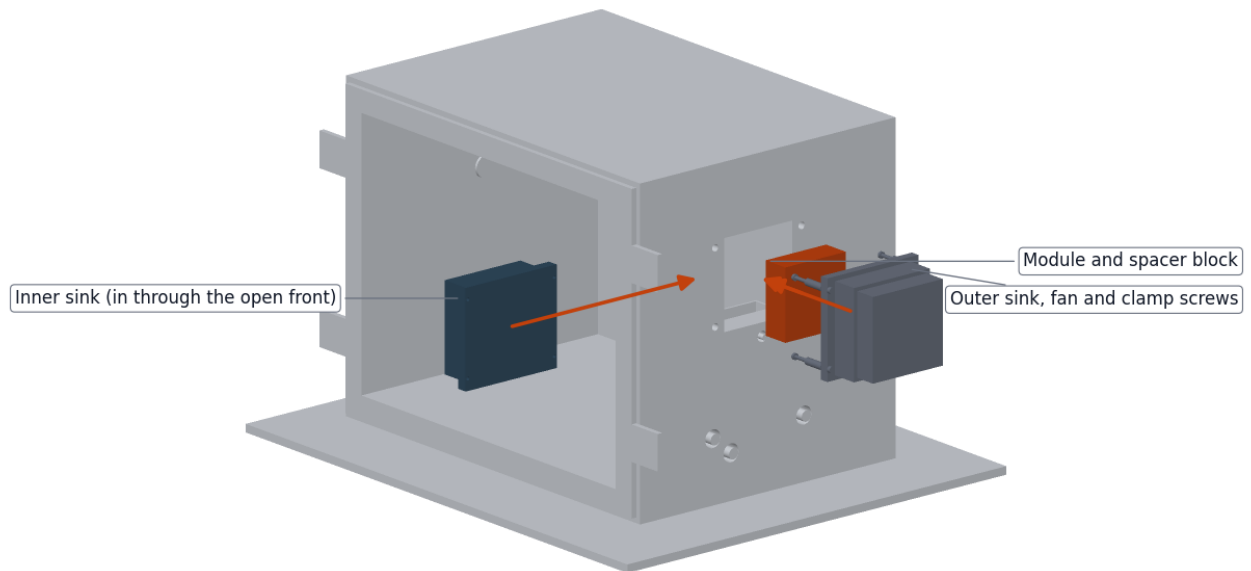
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Slide each panel over the fittings, glue it to the shell with foam-safe adhesive and tape every seam with foil tape. Push the clamp sleeves into the right panel's four small holes. Fill round each fitting with foam offcut.

Step 5: heat pump through the right wall

Step 5: Peltier assembly through the right wall

Thermal paste on the module; four clamp screws through the sleeves, tightened evenly in a cross pattern



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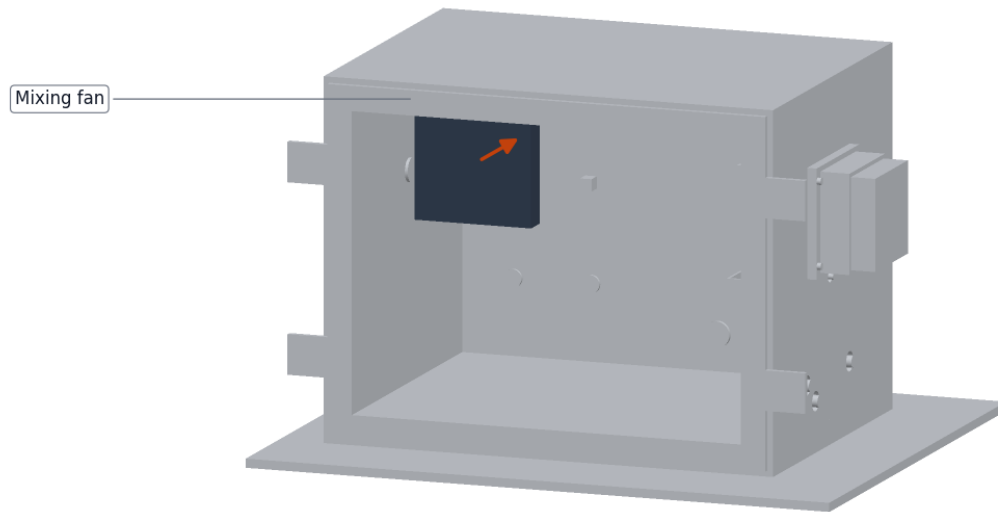
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Inner sink in through the open front onto its gasket; module and spacer block into the opening with thermal paste; outer sink and fan on the outside; four clamp screws through both bases, the wall and the sleeves, tightened evenly in a cross pattern. Seal the gap round the block. **Hold point:** safety stop S2 before the heat pump is ever powered.

Step 6: mixing fan onto its spacers

Step 6: mixing fan onto its spacers

Through the open front; four M4 nylon screws into the spacers, blowing toward the door



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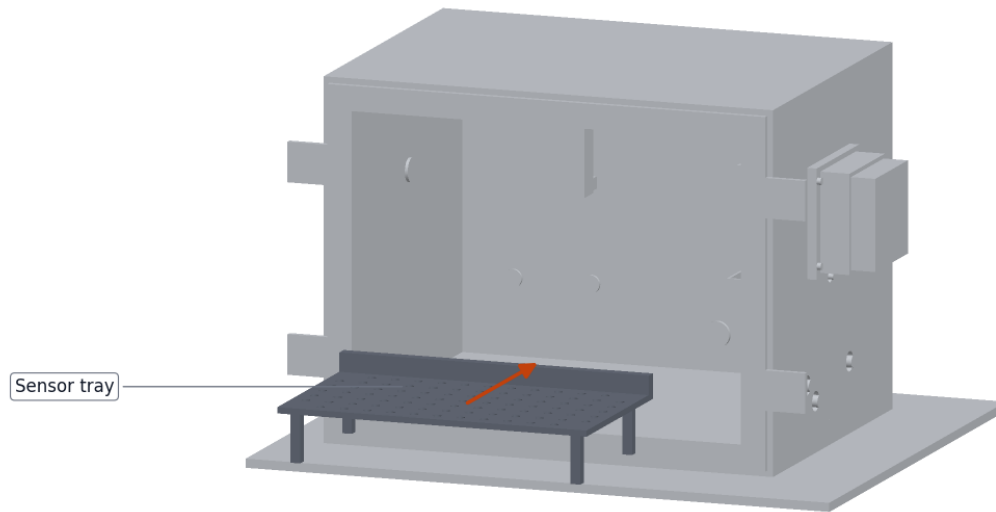
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Through the open front; four M4 nylon screws into the spacers, the fan blowing toward the door. Run its lead down the back wall to a gland.

Step 7: sensor tray in

Step 7: sensor tray in

Slide it in through the open front, 10 mm from the left wall and 16 mm behind the front opening; legs on the floor



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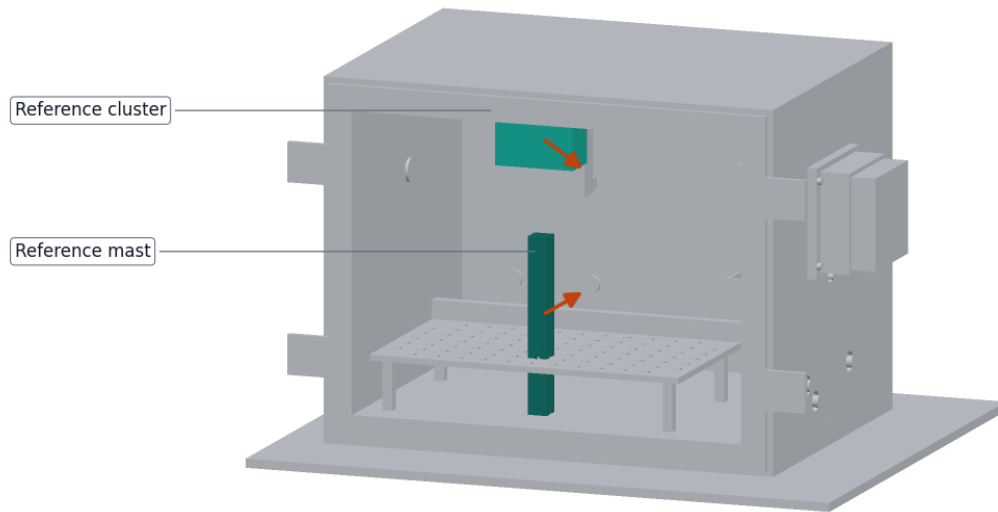
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Slide it in through the open front, 10 mm from the left wall and 16 mm behind the front opening, legs on the floor.

Step 8: reference mast and cluster

Step 8: reference mast and cluster

Mast against the back of the rail on two M4 nylon screws; cluster on its top; leads down the mast to the glands



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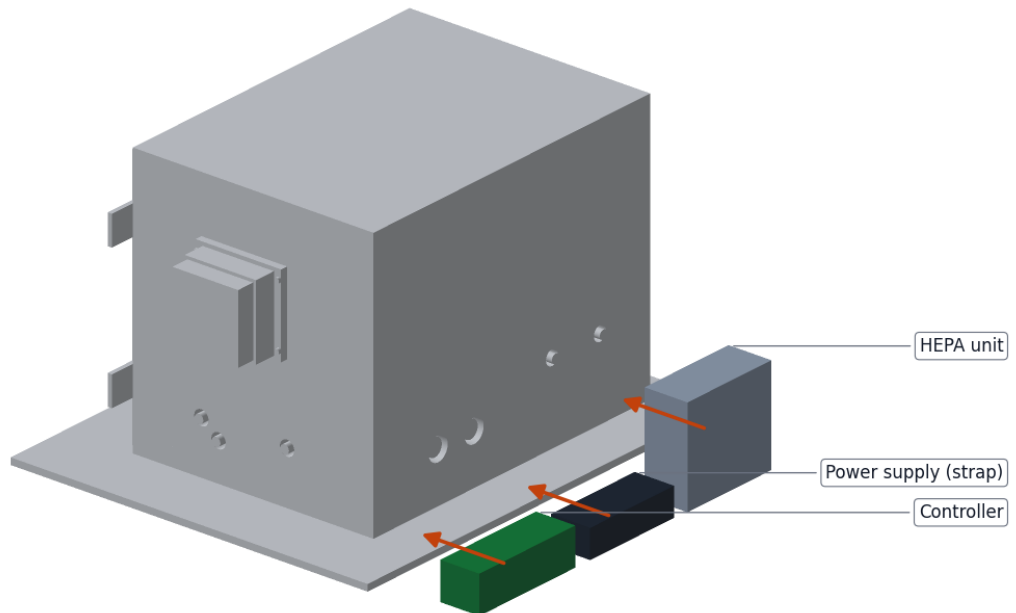
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Mast against the back of the rail on two M4 nylon screws; carrier on its top. Run the reference and sensor leads through the multi-hole inserts of the glands and tighten the gland caps.

Step 9: HEPA unit, controller and power supply onto the base

Step 9: HEPA unit, controller and power supply onto the base

HEPA unit pushed onto its two bulkheads, then screwed down; supply held by a strap. Seen from behind right



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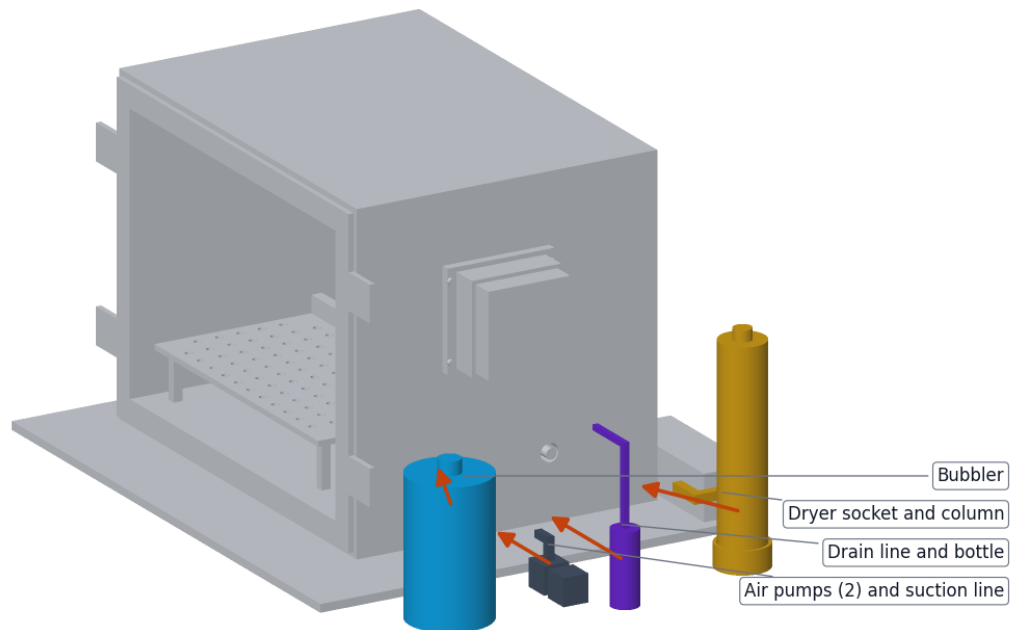
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Push the HEPA unit onto its two fittings, clamp the joints and screw it down; screw the controller down; strap the supply down. Seen from behind and to the right. Wire as Figure 26, with the fuse out.

Step 10: pumps, bubbler, dryer and drain bottle

Step 10: pumps, bubbler, dryer and drain bottle

Pumps screwed down; dryer socket screwed down; lines pushed onto the bulkheads and clamped



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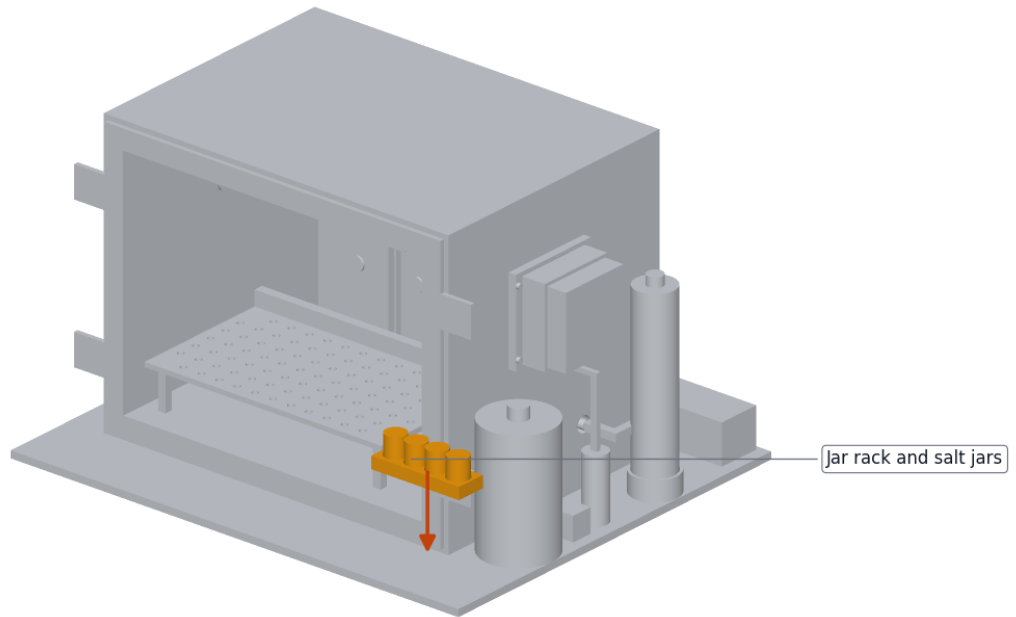
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Screw the pumps and the dryer socket down; set the bubbler and the bottle on their pads; push each line onto its fitting and clamp it, as Figure 25. The bubbler stays empty until safety stop S3.

Step 11: jar rack and salt jars

Step 11: jar rack and salt jars

Rack screwed to the base in front of the bubbler; jars stand in it, lids on



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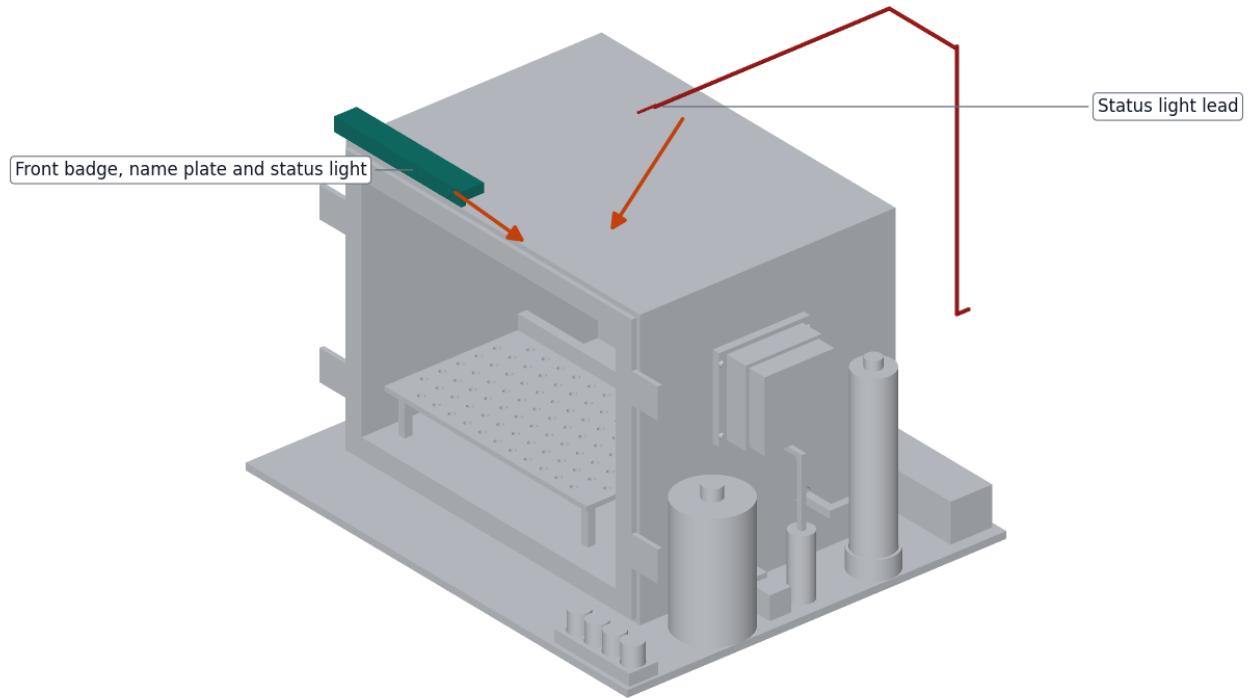
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Screw the rack down in front of the bubbler; stand the four jars in it with their lids on.

Step 12: front badge and status light lead

Step 12: front badge and status light lead

Badge glued to the top panel, lip on the frame's top edge; lead back along the top and down to the controller



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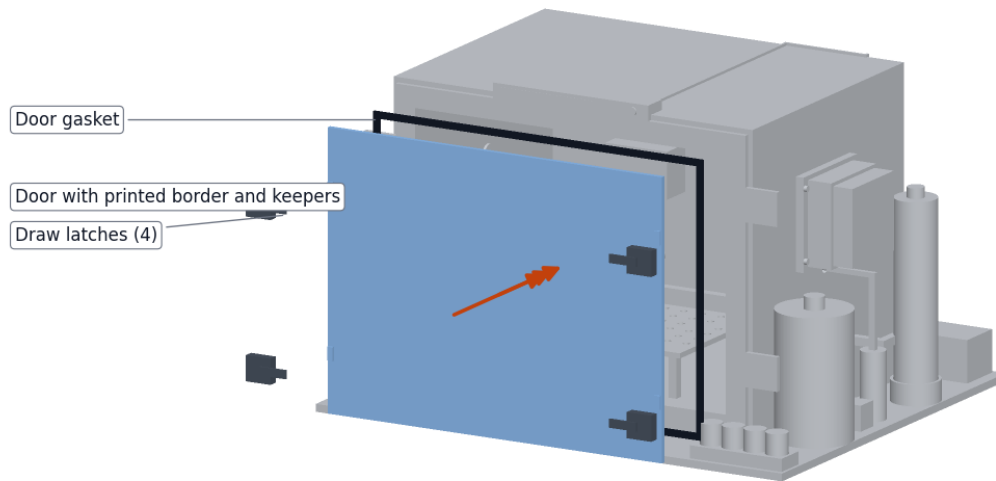
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Glue the badge to the front of the top panel with its lip on the front frame's top edge (section 3.15). Run the lead back along the top, along the back edge and down the back panel, clipped every 100 mm, and plug it into the controller.

Step 13: gasket, door and latches

Step 13: gasket, door and latches

Gasket on the frame face; door on the gasket; four latches screwed to the tabs, then closed



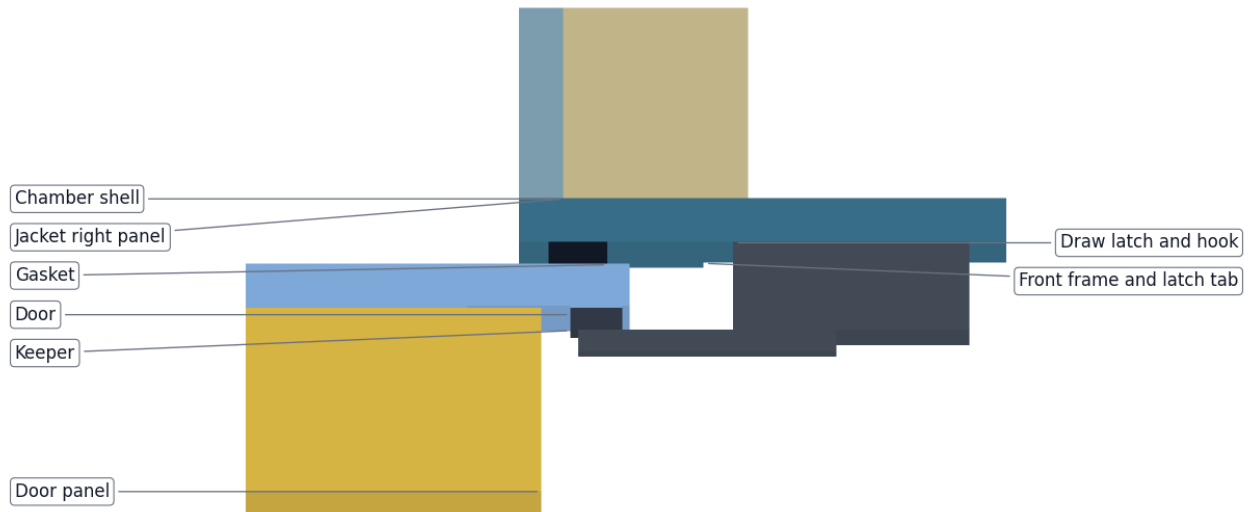
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Stick the gasket on the frame's face, 4 to 12 mm outside the opening, with its joint at the bottom. Screw the four latches to the frame tabs, hold the door on the gasket and close the latches.

Joint 9: door, gasket and latch (cut level with the upper right latch)

Seen from above. The latch on the frame tab hooks the keeper and squeezes the gasket



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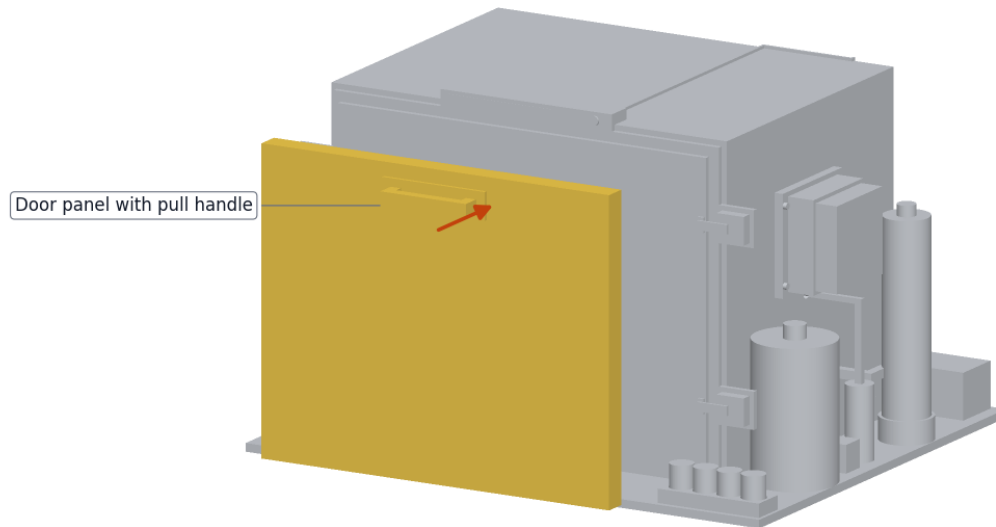
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Figure 27. Cut level with the upper right latch: the latch on the frame tab hooks the keeper on the door and squeezes the gasket between door and frame.

Step 14: door panel

Step 14: door panel (hot, humid and cold points)

Hold it by its pull handle and press it onto the four hook-and-loop pads on the door, between the latches



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Hold the panel by its pull handle and press it onto the four hook-and-loop pads on the door, between the latches. It is fitted for the 40 °C, 85 % RH point and for cold points, and off for the others.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of CLR-REQ-001.

Table 3. First checks.

Check	Requirement	How	Pass when
Shell leak	R2, R7	10 mm of water in the shell for an hour (section 3.3), before the jacket goes on	No water at any weld
Door seal	R2, R7	Paper strip trapped at 20 points round the closed door	Held tight at every point
Fit of the heads	R10	Six 90 x 70 x 50 mm blocks in the bays, door closed; leads through the glands	Everything fits; the gland caps seal round the leads
Fit of large heads	R10	A 192 x 152 x 234 mm block on four bays and a 192 x 70 x 234 mm block on two bays, in each position, door closed	Each stands on the tray clear of the walls, fan, inner sink and references
Status light	R15	Supply on, controller switching the light on and off	The light follows the controller; the lead stays clipped down
Fuse and supply	R15	Supply on, fuse in, every load off, then the heat pump at full drive with all fans, pumps and heaters on	Bus current 7.5 A or less; no part warm but the loads
Cut-offs	R16	Each bimetal switch warmed in a water bath with a thermometer	The relay drops out at 50 °C and at 70 °C (within the switch's tolerance) and the heat pump and heaters lose power
Heat pump direction	R1	Short runs at low drive in each direction	The inner sink warms when heating and cools when cooling; the outer fan runs whenever the module is powered
Air loops	R2	Each pump and the HEPA blower in turn, a soap film on each joint	Bubbles leave only where Figure 25 says air goes; no leak at any barb
Drain	R2	20 mL of water poured into the drip tray	All of it reaches the bottle
Clean-down	R7	HEPA unit at full speed, references logging	Particle reading falls below 2 µg/m ³
Size and mass	R13	Tape measure; bathroom scale with and without the rig held	Within 600 x 500 mm and 400 mm high (383 mm estimated); 14 kg or less (13.9 kg estimated)

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before cutting and welding acrylic.** Laser cutting in a cutter with working extraction; solvent cement used in a ventilated space with gloves and eye protection; no flame near the cement.
- **S2. Before the heat pump is powered.** Both sinks clamped flat with paste on the module; the outer fan runs whenever the module is powered (wired so it cannot be switched off while the module is on); both cut-offs wired in series with the relay coil and checked as in section 5; the 10 A fuse in place; the supply is certified and undamaged.
- **S3. Before the bubbler is filled and its heater powered.** Thermistors fitted to the pad and the line trace; the cut-off relay feeds the heater; the jar stands in its foam sleeve on its pad; the electronics stand behind the chamber, higher than any spilled water can reach. Fill with distilled water only.

- **S4. Before the first smoke run.** The room is ventilated; the incense cup stands outside the chamber on a non-combustible surface; the door is closed and latched; the HEPA unit is running. The door opens only after the references read below $5 \mu\text{g}/\text{m}^3$.
- **S5. Before handling soda lime or the salt slurries.** Gloves and eye protection on; containers labelled; children kept away. Exhaled-air carbon dioxide checks use a single-user bag and filter.
- **S6. Before the first unattended run.** One full sweep has been watched from start to finish, with the outer sink staying below 60°C and the chamber air below 45°C ; the rig stands on a non-combustible bench away from paper and curtains.

■ 7. Tools, skills and workspace

Tools. Laser cutter able to cut 6 mm cast acrylic (a maker space or a cutting service) or a fine-toothed saw and a router; acrylic solvent cement with a needle applicator; corner clamps and square blocks; fine-toothed saw or hot-wire cutter for foam board; drill with 3, 4.2 and 4.5 mm bits and a step drill to 20 mm; M3 and M4 taps; 3D printer able to print PETG (bed at least 120×60 mm, height 180 mm); screwdrivers and nut drivers; side cutters, wire strippers and a ferrule crimper; soldering iron; multimeter; digital thermometer; tape measure, steel rule and square; bathroom scale.

Skills. No certified trade is needed. Marking out, laser cutting or sawing acrylic, solvent welding, cutting foam board, drilling and tapping, 3D printing, crimping and soldering, and safe use of a 12 V supply. Everything runs at 12 V from a certified external supply; no mains wiring is part of this build.

Workspace. A bench about 1.2×0.8 m in a ventilated room; a flat surface to weld the shell square; a separate corner for solvent work; a sink for the leak check.

Personal protective equipment. Safety glasses for cutting, drilling, welding and soldering; nitrile gloves for solvent cement, soda lime and salts; a dust mask when sanding acrylic.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 116 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/CLR-DWG-101` to `CLR-DWG-112` .
- General arrangement: `cad/drawings/CLR-DWG-001.pdf` , Rev P5.
- Calculations: `docs/04-calcs/01-sizing.md` (CLR-CAL-001 v0.7) and `docs/04-calcs/sizing.py` ; mass [A3], large heads [A4], heat capacity and ramps [B7, B8], power [J1], cut-offs [J2], cost [K1].
- Bill of materials: `bom/bom.csv` and `bom/bom-notes.md` .
- Decisions: `docs/decisions/0003-design-for-construction.md` (CLR-DDR-003), with CLR-DDR-001 and CLR-DDR-002; decisions and items to confirm in `docs/06-design-decisions.md` (CLR-DEC-001).
- Requirements: `docs/03-requirements.md` (CLR-REQ-001 v0.9).